Fractal, Self-similarity and Power law

Language of Complex System

0905/2025 Ruyi Tao

- What is Fractal — Self-Similarity
  - Origin of Fractals: Mathematics Monsters
  - Definition and Quantification of Fractals
    - Definition Proposed by Mandelbrot
    - Proposition and Calculation of Fractal Dimensions

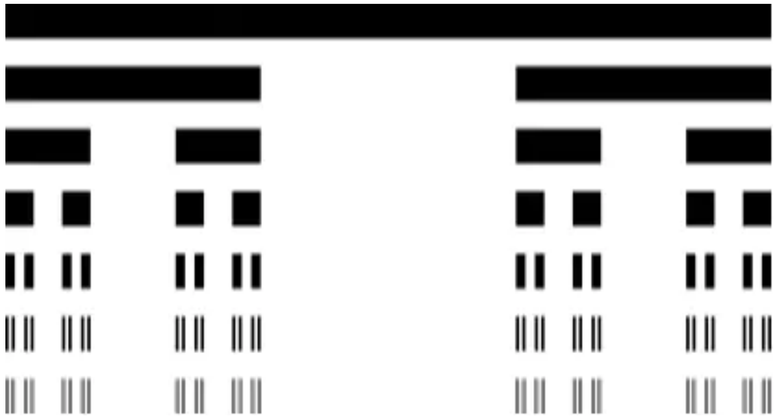
- "Fractals" in Complex Systems — Scaling Theory
  - Kleiber's Law and Its Origin
  - Urban Scaling Laws and Their Origin
  - Underlying Universal Mechanisms

- Broader Discussion of Self-Similarity in Complex Systems
  - Critical Phase Transition and Self-Similarity
  - Self-Similarity in Dynamics
  - Self-Similarity in Complex Networks

Origin of Fractals: The "Monsters" in Mathematics

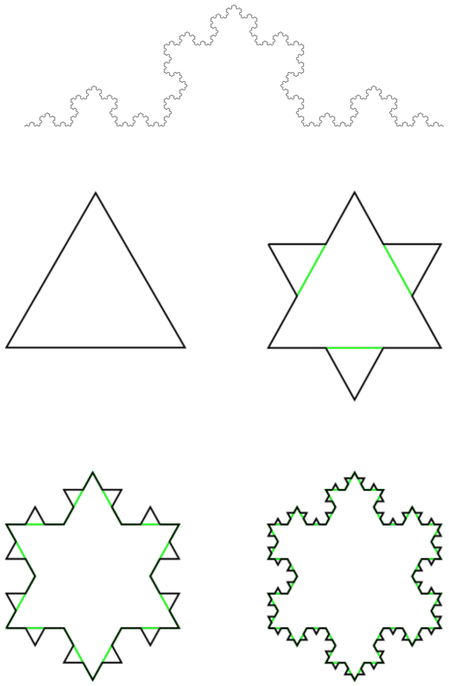
Cantor Set (1875)

As numerous as the real numbers, yet with measure zero



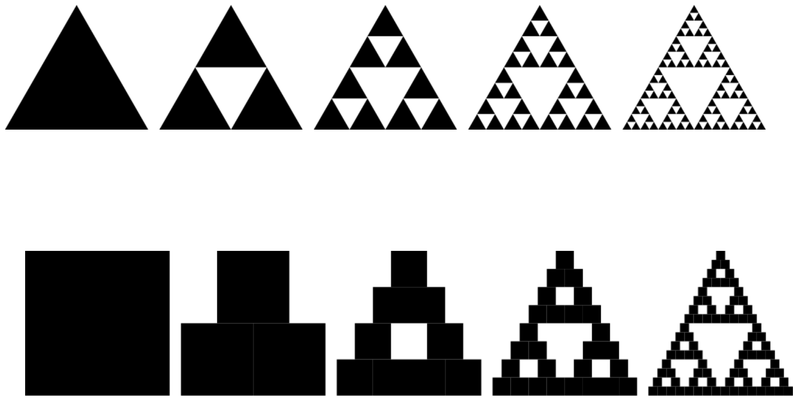
Koch Curve & Koch Snowflake (1904)

Finite area but infinite perimeter

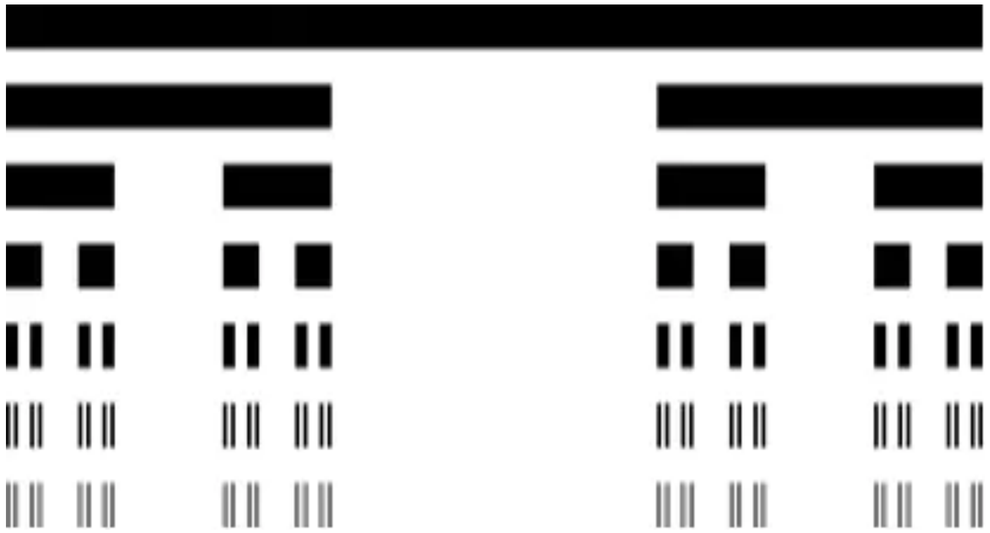


Sierpiński Triangle

Zero area



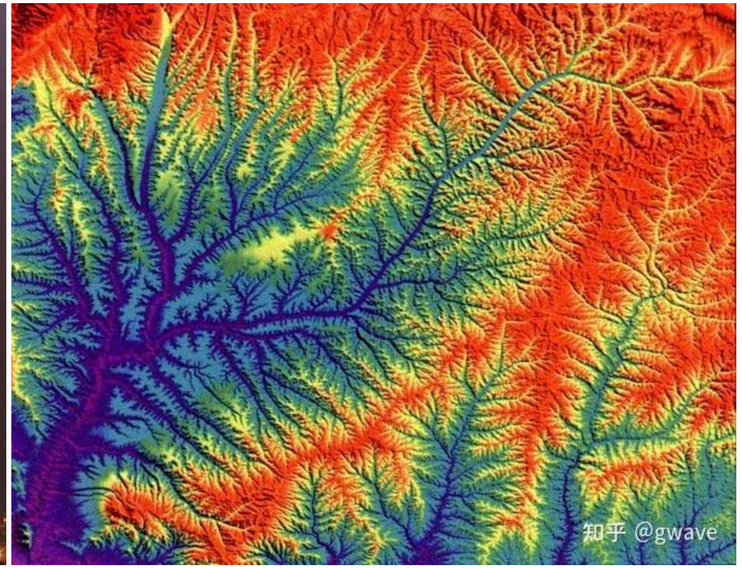
Origin of Fractals: The "Monsters" in Mathematics



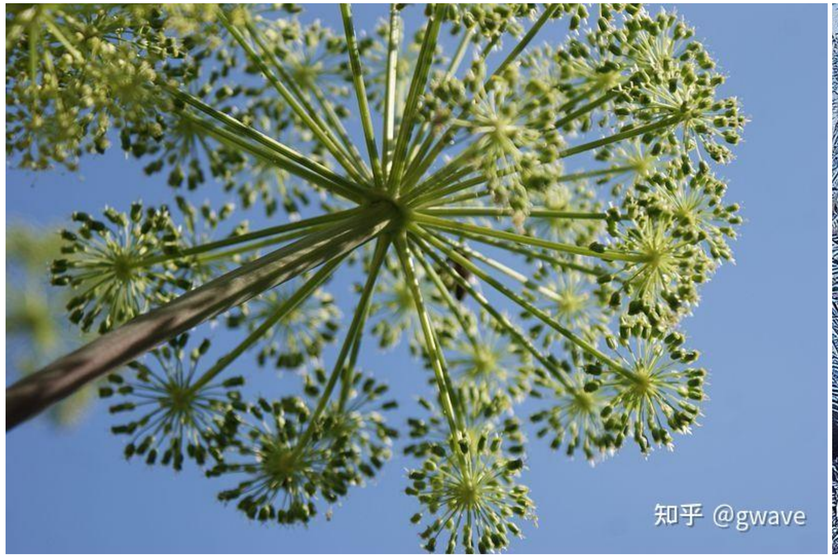
For a Cantor Set C

- By repeatedly magnifying part of C, we can continuously observe complex structures fragmented across multiple scales; this structure is endless.
- It contains many smaller figures that are self-similar to itself at different scales.

Fractals in Nature

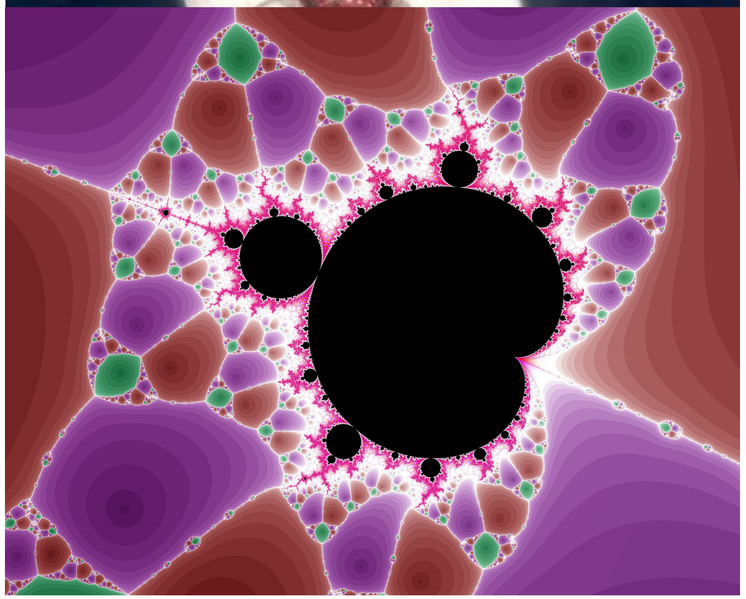


"A realistic statement is that most curves encountered in nature are non-rectifiable."



Mandelbrot and Mandelbrot Set — Formal Proposition of Fractal Concept

- Mandelbrot: Scientist at IBM
- In the 1960s，he began researching rough structures in nature
- In 1975, he formally proposed the term “fractal,” defining it as “a geometrical object with self-similar structure and non-integer dimension.”
- In 1980, he published The Fractal Geometry of Nature, establishing the framework of fractal geometry.



How Long Is the Coast of Britain?

Statistical Self-Similarity and Fractional Dimension

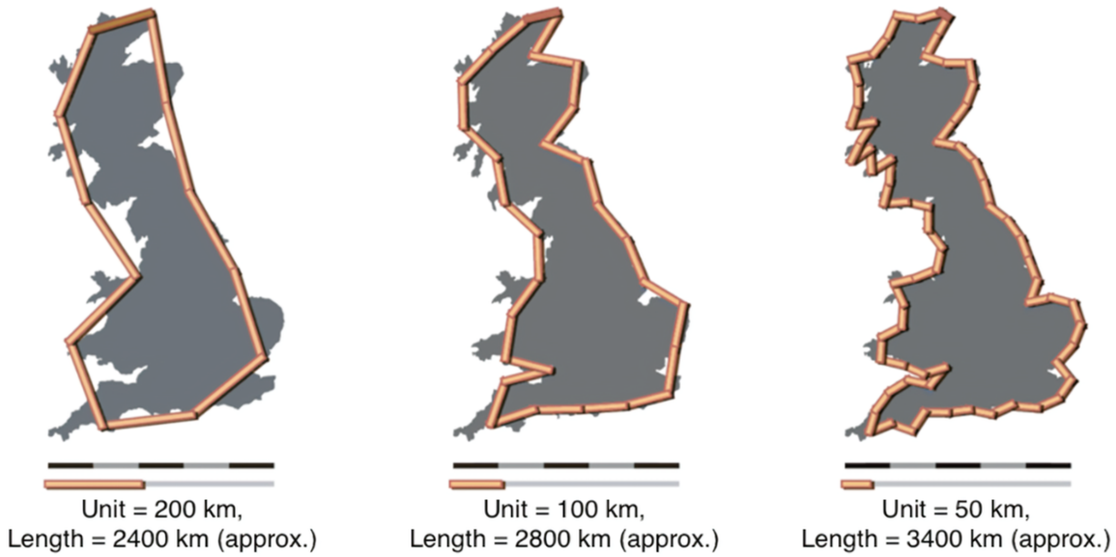
Abstract. *Geographical curves are so involved in their detail that their lengths are often infinite or, rather, undefinable. However, many are statistically “self-similar,” meaning that each portion can be considered a reduced-scale image of the whole. In that case, the degree of complication can be described by a quantity D that has many properties of a “dimension,” though it is fractional; that is, it exceeds the value unity associated with the ordinary, rectifiable, curves.*

How to Measure a Coastline

Traditional method: Use a straight line segment of length G to “walk” along the coastline on a map, ensuring that the endpoints of each segment lie on the coastline, and count the number of steps. This gives an estimated length denoted by L(G)

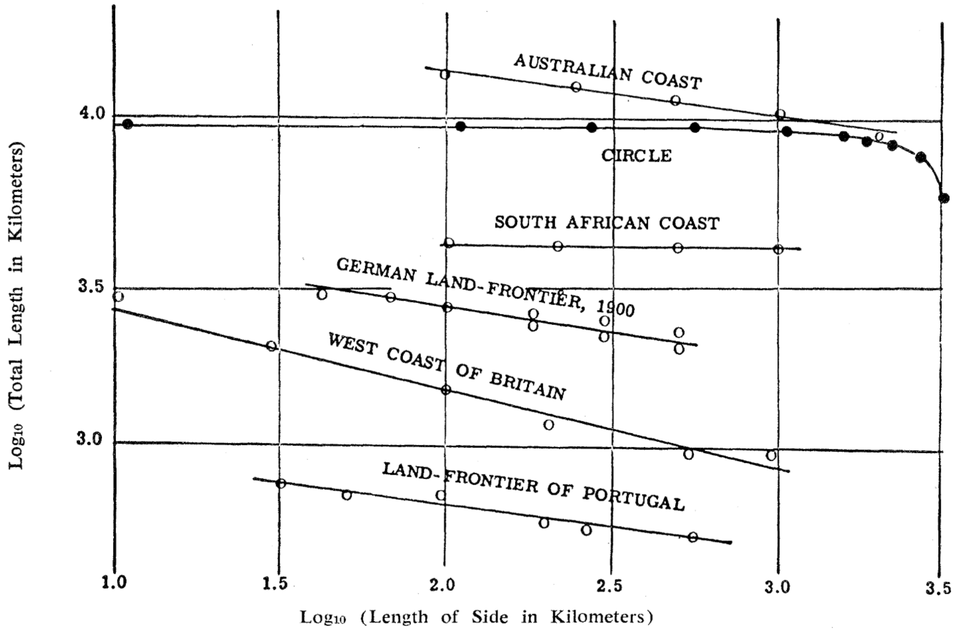
How to Measure a Coastline

- L(G) is highly dependent on the choice of G.
  It is necessary to select multiple Gs to obtain different L(G).



Lewis F. Richardson proposed the empirical formula, where M is a positive constant and D is a constant no smaller than 1.:

$$L(G) = MG^{1-D}$$



The more convoluted the coastline, the larger the value of D.
The west coast of Great Britain has one of the largest D values, approximately 1.25.

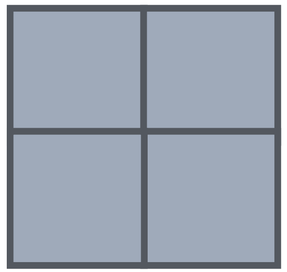
Benoit Mandelbrot. (1967). How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension. Science, 156(3775), 636–638.

How to Measure a Coastline

Total length is N

The total length of the subsegments is 1/r of the total length.

Dimension=$\ln N / \ln r = 1$



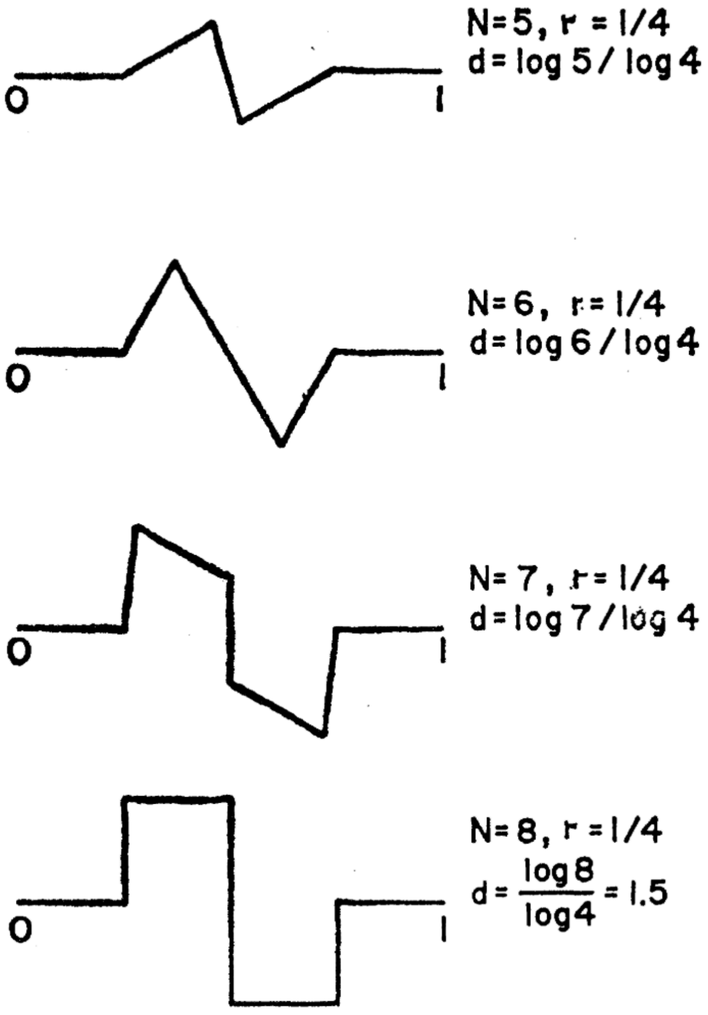
Total area is N

The sub-area is $1/r^2$ of the total area

Dimension=$\ln N / \ln r = 2$

Benoit Mandelbrot. (1967). How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension. Science, 156(3775), 636–638.

How to Measure a Coastline



Each subpart is obtained from the whole figure through a similarity transformation with scaling factor r.

Definition of dimension D:

$$D = -\ln N / \ln r$$

Benoit Mandelbrot. (1967). How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension. Science, 156(3775), 636–638.

Statistical Self-Similarity

Definition of Random Figures

- Set of possible figures: Ω； Assign a probability to each figure.

Conditions for Self-Similarity

1.Each figure is composed of N subfigures.

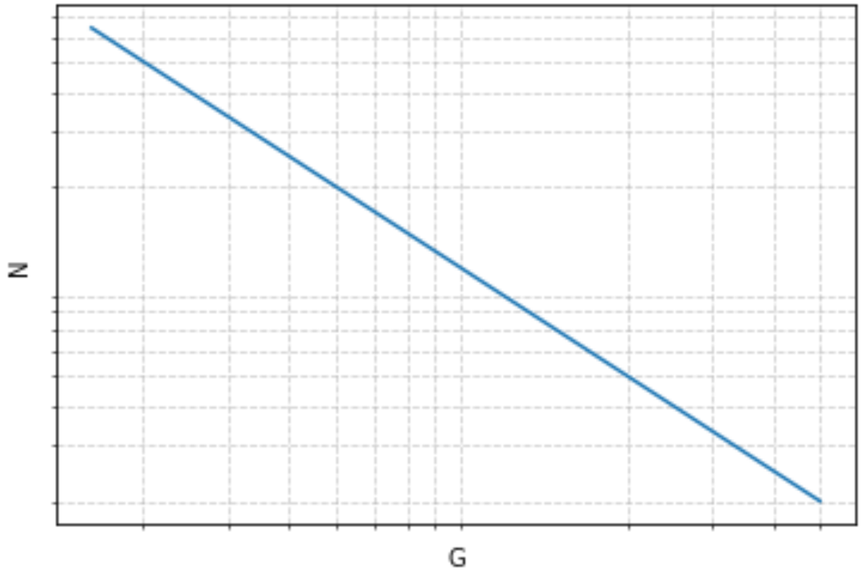
- ach subfigure is a similar scaling (ratio r) of the whole.
- Exact overlap is not required; only statistical regularities such as mean and fluctuation patterns need to be consistent.

2.Probability consistency:：

- The probability distribution obtained by “selecting the whole” is consistent with that obtained by “assembling from subfigures.”

Fractal dimension： D=−logN/logr

$$N(G) \sim G^{1-D}$$

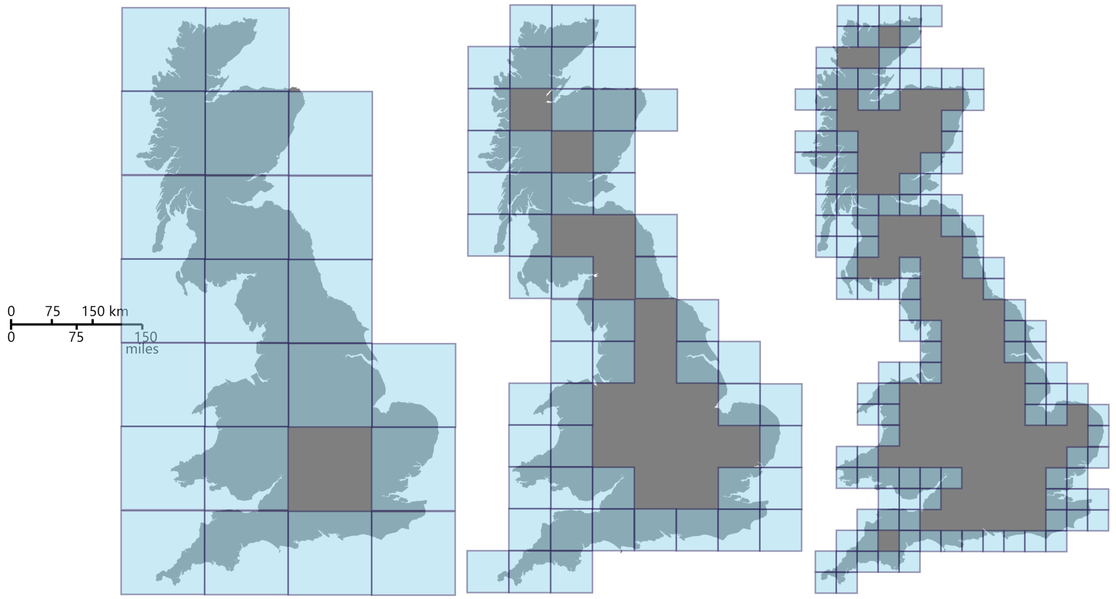


Benoit Mandelbrot. (1967). How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension. Science, 156(3775), 636–638.

Measurement of Fractal Dimension

Box-counting (Minkowski Dimension)

1. To calculate the dimension of a fractal S, imagine placing it on a uniformly divided grid and counting the minimum number of boxes needed to cover the fractal.
2. By progressively refining the grid and examining how the required number of covering boxes changes, the box-counting dimension can be computed.



$$\dim_{\text{box}}(S) := \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}.$$

Summary

Fractals arise as an extension of Euclidean geometric dimensions.

They are applied to the measurement of the real world.

Power laws serve as signals revealing the self-similarity of geometric structures.

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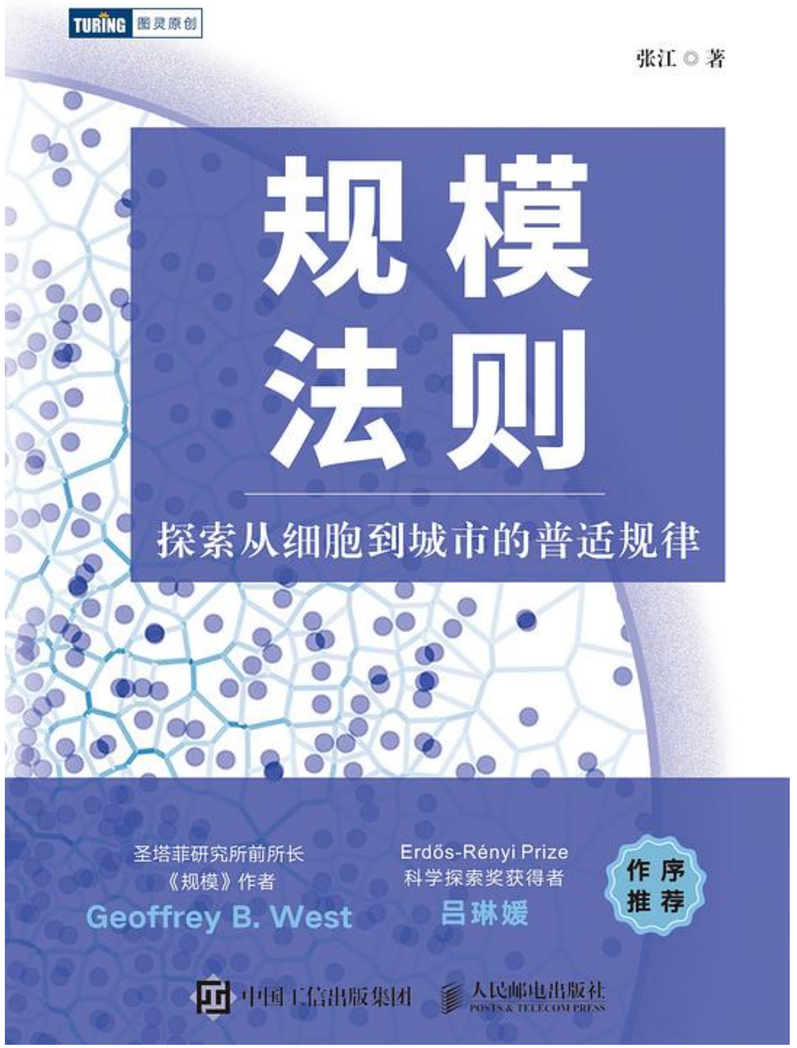
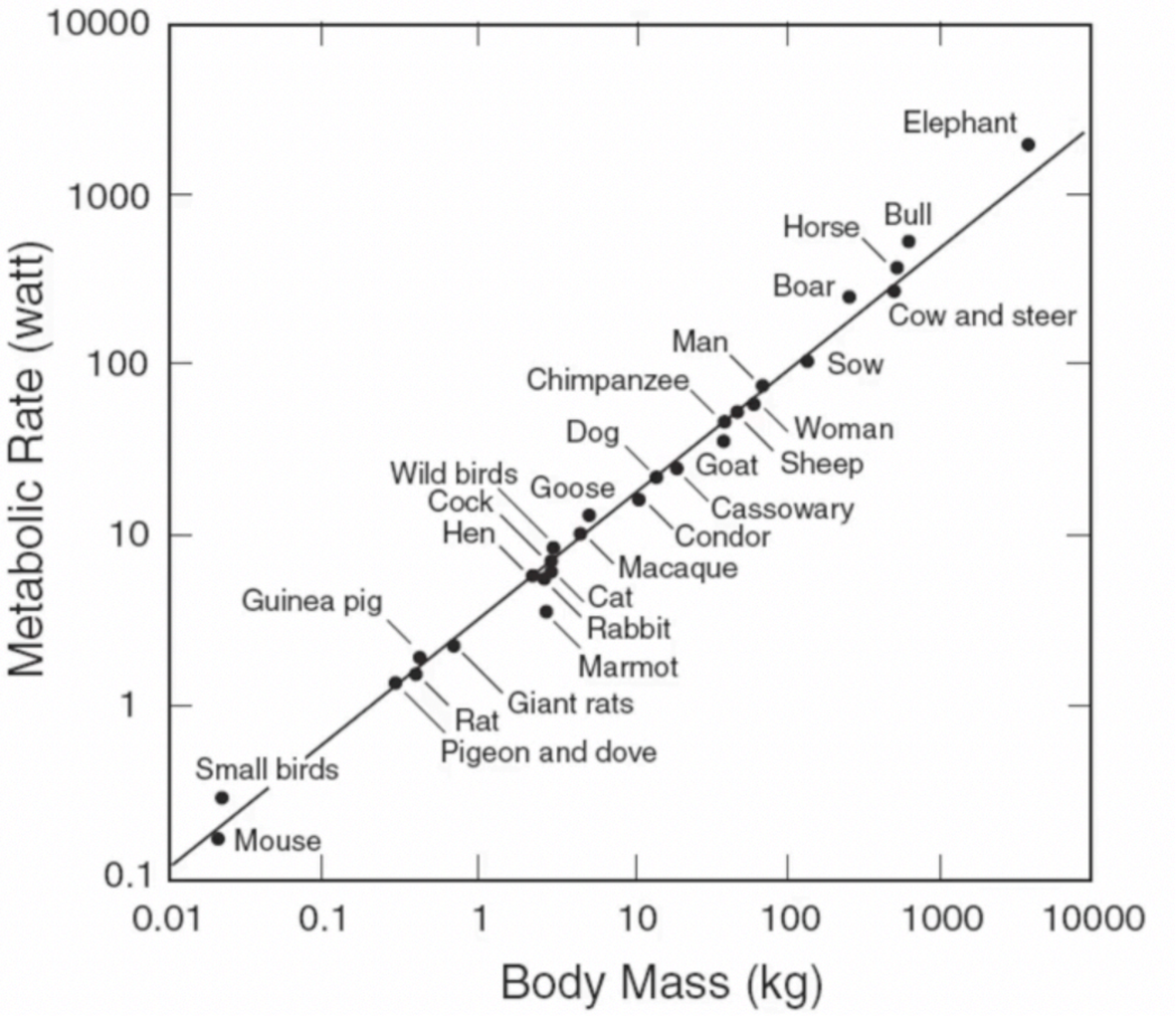
Scaling Theory

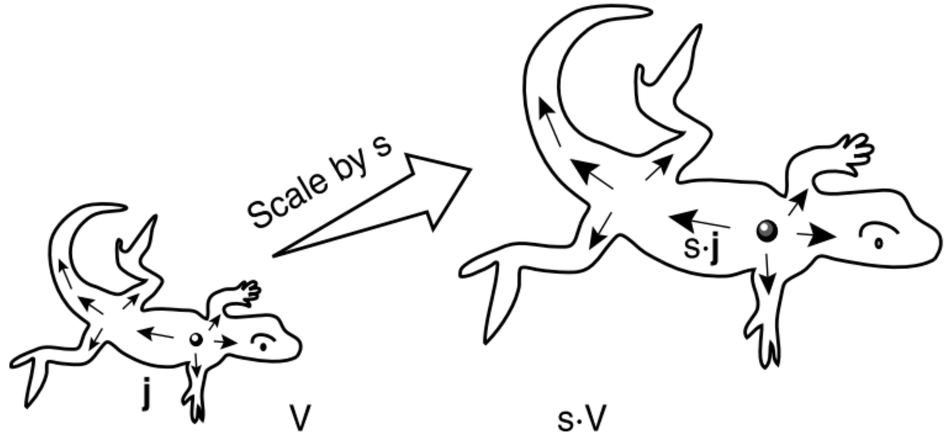


Kleiber's Law

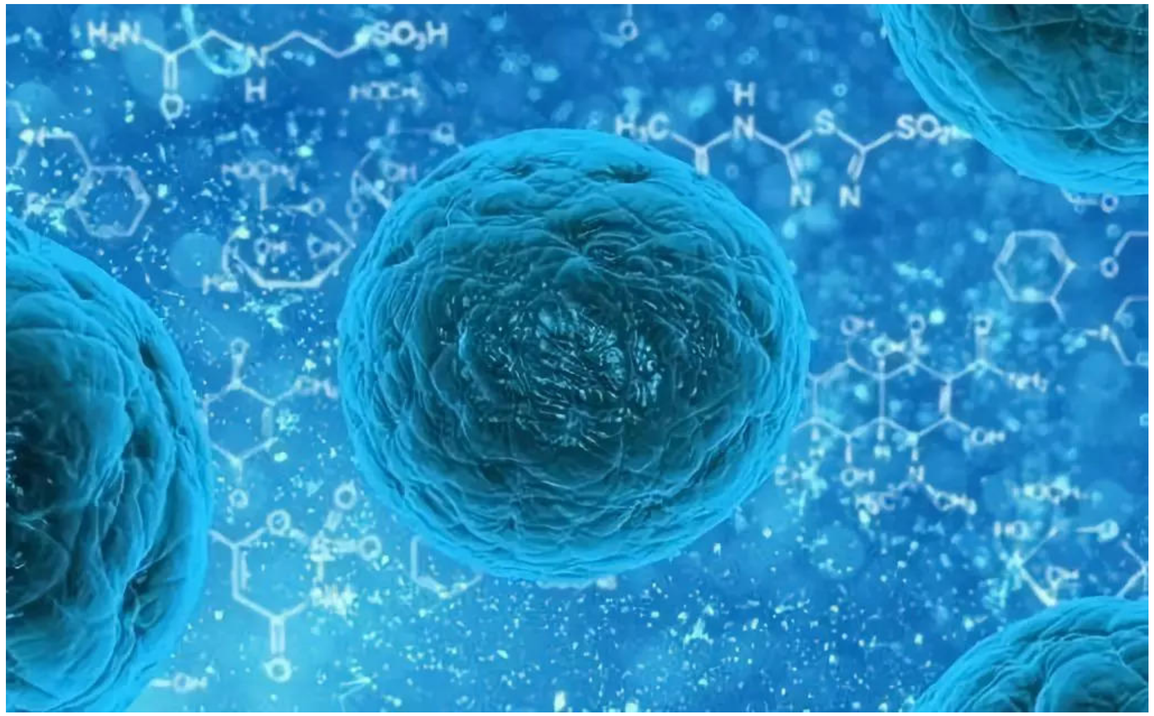


Relationship between metabolic rate and body mass.

$$F = cM^{3/4}$$



Origins of Kleiber's Law



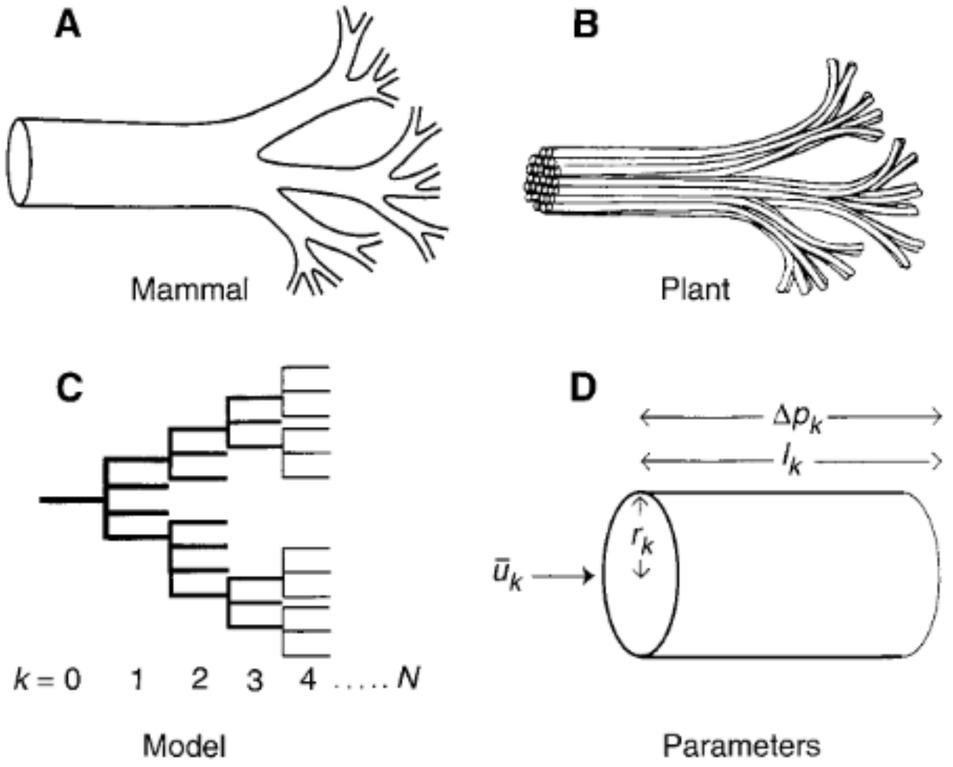
- Your lungs occupy only about the volume of a soccer ball, around 5–6 liters, but the total surface area of the alveoli responsible for oxygen and carbon dioxide exchange is nearly as large as a tennis court.
- The total length of all airflow pathways is nearly the distance from London to Moscow, about 2500 kilometers.
- Fractal structures maximize the utilization of blood vessels and cellular surface area for transporting matter and energy.

Origins of Kleiber's Law — The WBE Model

Assumptions

1. The transport network is fractal. Each layer fills the entire space, forming the volume/mass of the organism.
2. The capacity (radius) of the terminal branches does not vary with system size.
3. Energy loss in transport is minimized — cross-sectional area is conserved across layers.

4.



流量是代谢，体积是质量

West, G. B., Brown, J. H., & Enquist, B. J. (1997). A general model for the origin of allometric scaling laws in biology. Science, 276(5309), 122–126.

Origins of Kleiber's Law — The WBE Model

A general model for the structure and allometry of plant vascular systems

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A general quantitative theory of forest structure and dynamics

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Contributed by James H. Brown, December 5, 2008 (sent for review October 1, 2008)

Functional Ecology 2004
18, 283–289

FORUM

Is West, Brown and Enquist's model of allometric scaling mathematically correct and biologically relevant?

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Functional Ecology 2005
19, 735–738

FORUM

Yes, West, Brown and Enquist's model of allometric scaling is both mathematically correct and biologically relevant

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Functional Ecology 2005
19, 739–743

FORUM

West, Brown and Enquist's model of allometric scaling again: the same questions remain

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Scaling law in City

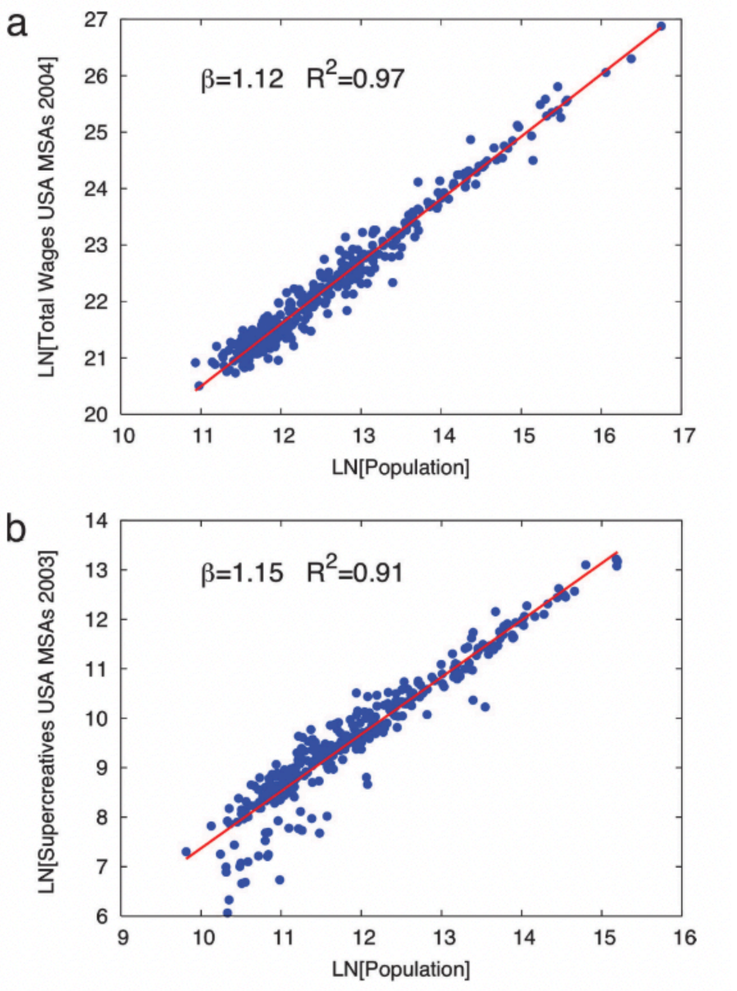


Table 1. Scaling exponents for urban indicators vs. city size

Y	β	95% CI	Adj-R^2	Observations	Country–year
New patents	1.27	[1.25,1.29]	0.72	331	U.S. 2001
Inventors	1.25	[1.22,1.27]	0.76	331	U.S. 2001
Private R&D employment	1.34	[1.29,1.39]	0.92	266	U.S. 2002
"Supercreative" employment	1.15	[1.11,1.18]	0.89	287	U.S. 2003
R&D establishments	1.19	[1.14,1.22]	0.77	287	U.S. 1997
R&D employment	1.26	[1.18,1.43]	0.93	295	China 2002
Total wages	1.12	[1.09,1.13]	0.96	361	U.S. 2002
Total bank deposits	1.08	[1.03,1.11]	0.91	267	U.S. 1996
GDP	1.15	[1.06,1.23]	0.96	295	China 2002
GDP	1.26	[1.09,1.46]	0.64	196	EU 1999–2003
GDP	1.13	[1.03,1.23]	0.94	37	Germany 2003
Total electrical consumption	1.07	[1.03,1.11]	0.88	392	Germany 2002
New AIDS cases	1.23	[1.18,1.29]	0.76	93	U.S. 2002–2003
Serious crimes	1.16	[1.11, 1.18]	0.89	287	U.S. 2003
Total housing	1.00	[0.99,1.01]	0.99	316	U.S. 1990
Total employment	1.01	[0.99,1.02]	0.98	331	U.S. 2001
Household electrical consumption	1.00	[0.94,1.06]	0.88	377	Germany 2002
Household electrical consumption	1.05	[0.89,1.22]	0.91	295	China 2002
Household water consumption	1.01	[0.89,1.11]	0.96	295	China 2002
Gasoline stations	0.77	[0.74,0.81]	0.93	318	U.S. 2001
Gasoline sales	0.79	[0.73,0.80]	0.94	318	U.S. 2001
Length of electrical cables	0.87	[0.82,0.92]	0.75	380	Germany 2002
Road surface	0.83	[0.74,0.92]	0.87	29	Germany 2002

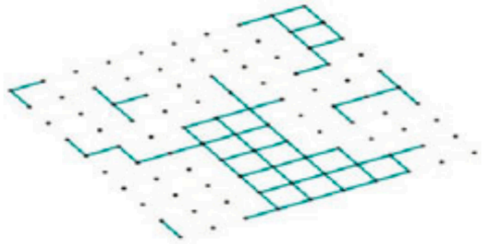
Data sources are shown in *SI Text*. CI, confidence interval; Adj-R^2, adjusted R^2; GDP, gross domestic product.

Three types of scaling exponents.

Bettencourt, L. M. A., Lobo, J., Helbing, D., Kühnert, C., & West, G. B. (2007). Growth, innovation, scaling, and the pace of life in cities. *PNAS*, *104*(17), 04

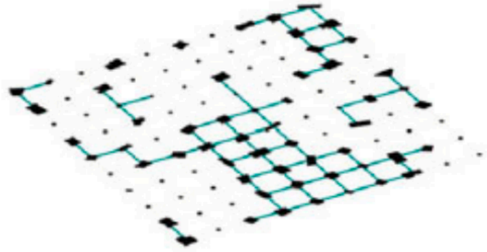
Scaling law in City

(a)



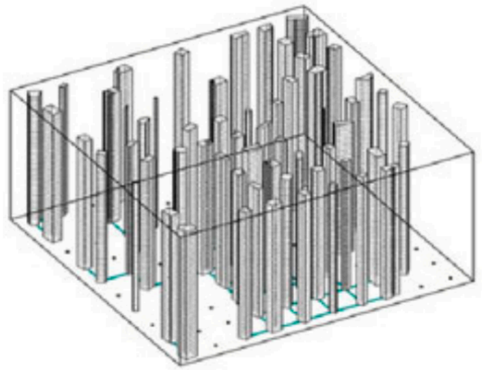
$\ell = L^{d_i}$

(b)



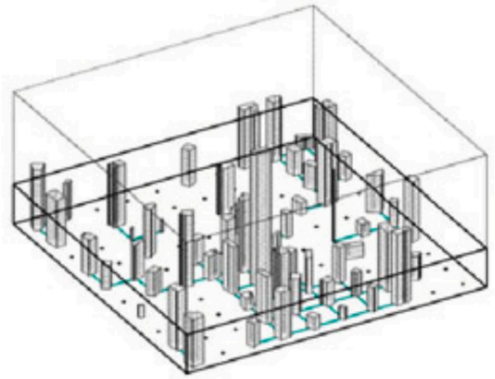
$L^{dp_p} = L^{d_i}$

(d)



$p = L^{dp} = L^{dp_p + 1}$

(e)

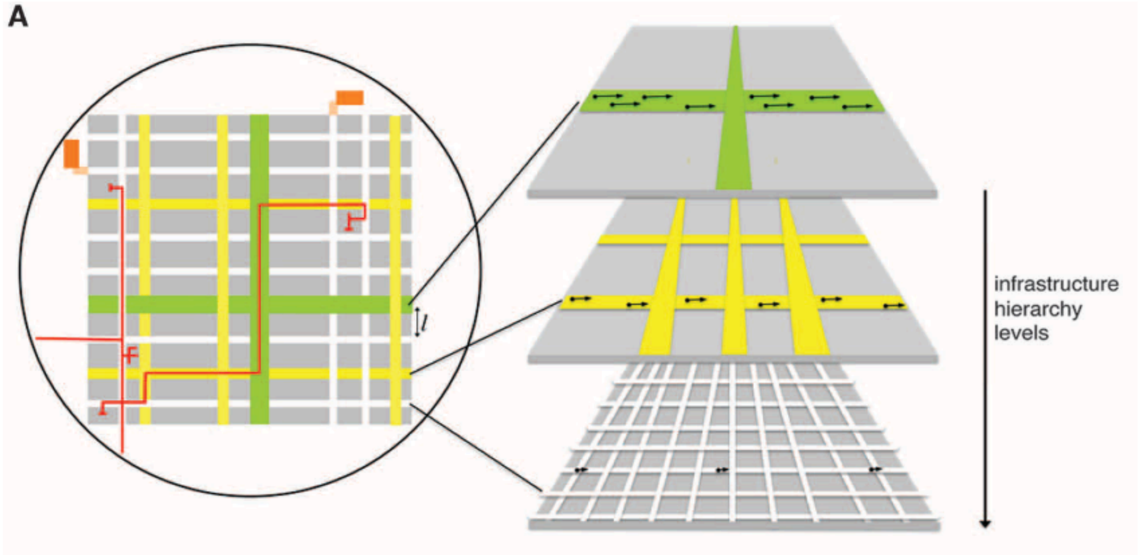


$p = L^{dp} = L^{dp_p + \eta}$

- Sublinear scaling arises from the ratio between the fractal dimension of road networks and the fractal dimension of three-dimensional population distribution.
- Superlinear scaling emerges from human interactions constrained by urban geometry.
- The sum of sublinear and superlinear exponents is close to 2.

Molinero, C., & Thurner, S. (2021). How the geometry of cities determines urban scaling laws. Journal of the Royal Society Interface

Origins of Urban Scaling Laws — Bettencourt Model



Urban scaling relations	Observed exponent range	Model ($D = 2$, $H = 1$)	Model D, H
Land area $A = aN^{\alpha}$	[0.56,1.04]	$\alpha = \frac{2}{3}$	$\alpha = \frac{D}{D+H}$
Network volume $A_n = A_0 N^{\nu}$	[0.74,0.92]	$\nu = \frac{5}{6}$	$\nu = 1 - \delta$
Network length $L_n = L_0 N^{\lambda}$	[0.55,0.78]	$\lambda = \frac{2}{3}$	$\lambda = \alpha$
Interactions per capita $\bar{I}_i = I_0 N^{\delta}$	[0.00,0.25]	$\delta = \frac{1}{6}$	$\delta = \frac{H}{D(D+H)}$
Socioeconomic rates $Y = Y_0 N^{\beta}$	[1.01,1.33]	$\beta = \frac{7}{6}$	$\beta = 1 + \delta$
Network power dissipation $W = W_0 N^{\omega}$	[1.05,1.17]	$\omega = \frac{7}{6}$	$\omega = 1 + \delta$
Average land rents $P_L = P_0 N^{\delta_L}$	[0.46,0.52]	$\delta_L = \frac{1}{2}$	$\delta_L = 1 - \alpha + \delta$

Four Principles

1. Population is fully mixed.
2. Road networks grow.
3. Human social capacity is limited: the total effort G each person spends on movement and interaction does not increase with city size.
4. Total urban output is proportional to the amount of social interaction.

Hierarchical Model Setup

- The road network has a fractal structure with h layers and branching ratio b.
- The tree contains b^h leaf nodes.
- Leaf nodes correspond to infrastructure units.
- Physical quantities associated with roads:
  - unit segment length at scale i: l_i
    - urban area covered: a_i

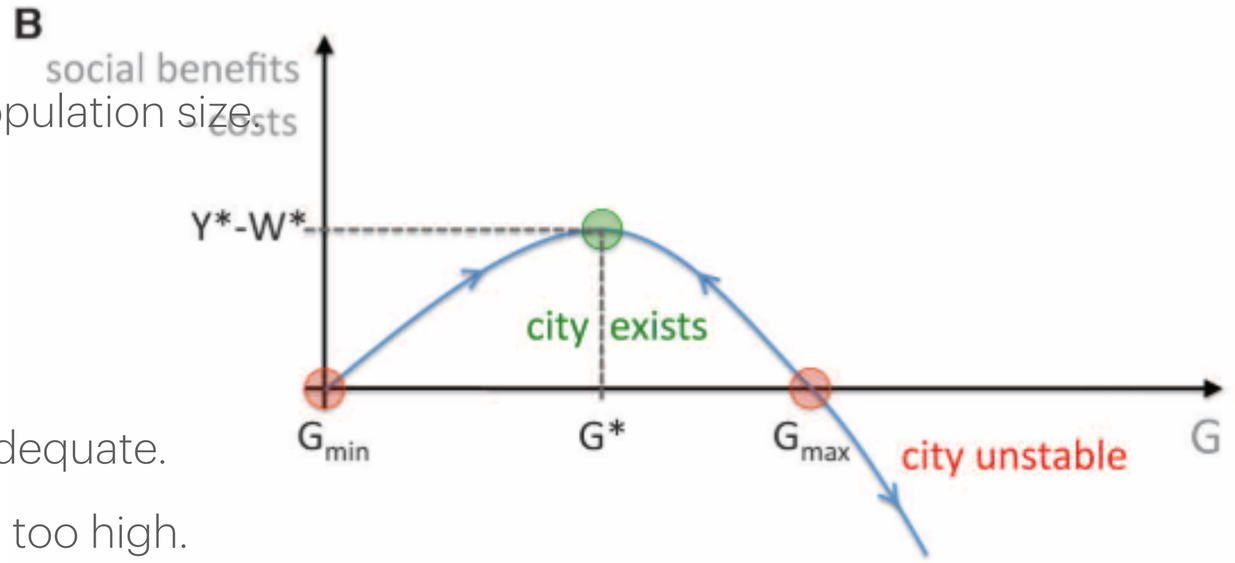
Bettencourt, L. M. A. (2013). The origins of scaling in cities. Science, 340(6139), 1438–1441. https://doi.org/10.1126/science.1235823

Origins of Urban Scaling Laws — Bettencourt Model

Calculate output Y and energy consumption W.

Ultimately:

- Y/W becomes a city-efficiency indicator independent of population size.
- Optimal urban planning is obtained by optimizing G.



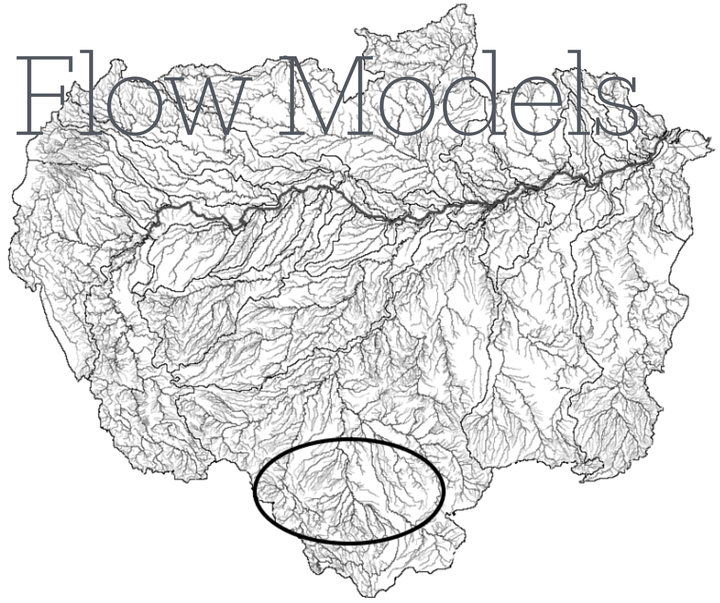
Core insight:

If G < G*, infrastructure is insufficient and interactions are inadequate.

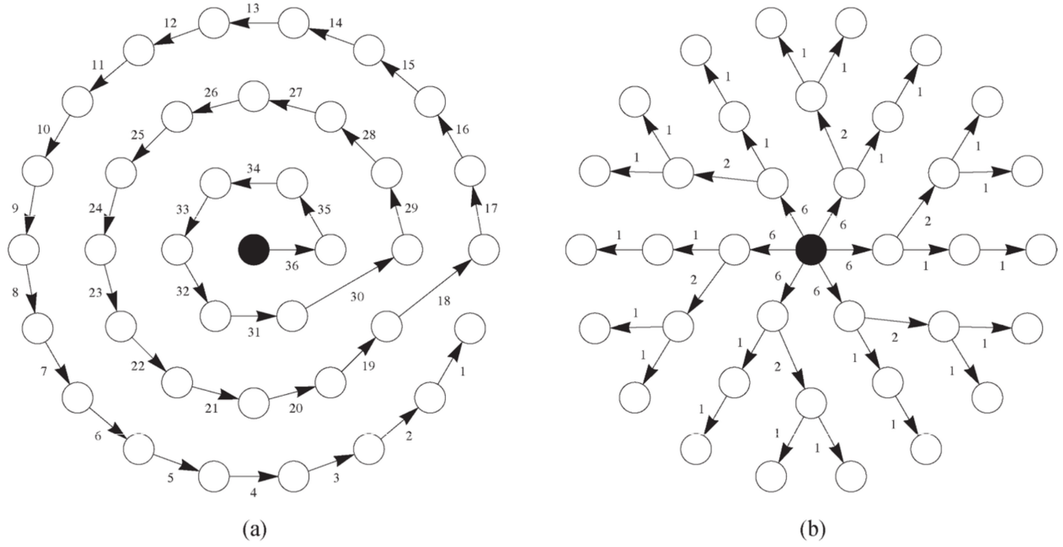
If G > G*, infrastructure investment is excessive and costs are too high.

Bettencourt, L. M. A. (2013). The origins of scaling in cities. Science, 340(6139), 1438–1441. https://doi.org/10.1126/science.1235823

Potential Universal Mechanism — Flow Models



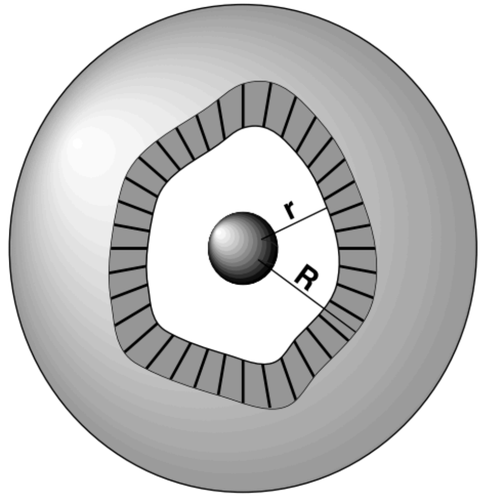
- Transport units are uniformly distributed in space and regarded as metabo to volume.
- Total flow corresponds to transport distance, representing cost; the systen
- Flow occurs locally.



Optimal Transport

Banavar, J. R., et al. (1999), Nature

$$F \sim M^{d/(d+1)}$$



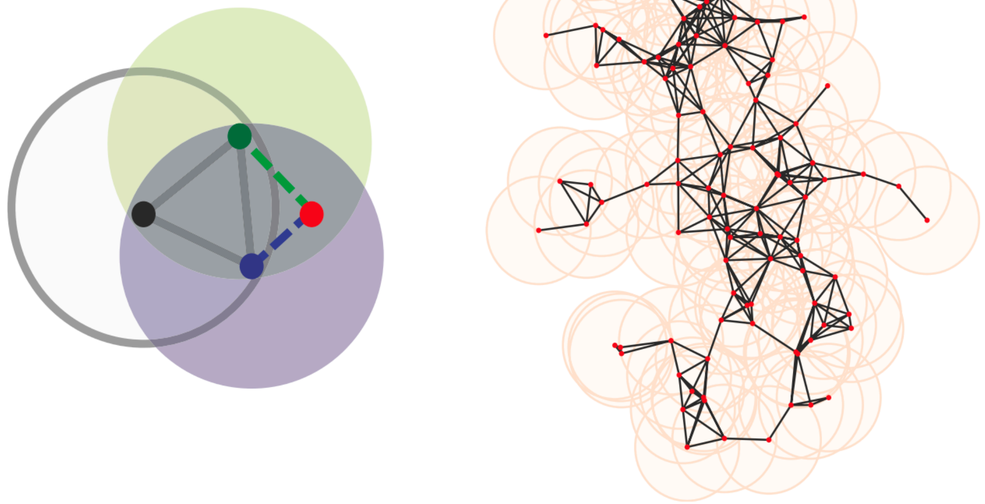
Continuous flow model:

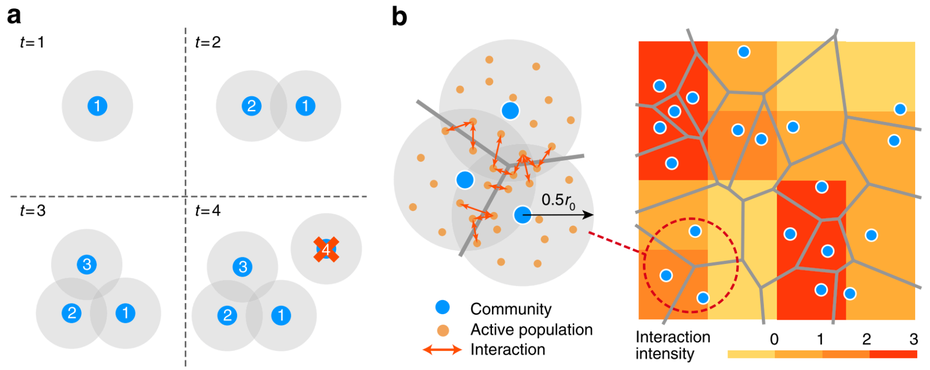
Dreyer, O. (2001).PRL

Potential Universal Mechanism — Flow Models

- Growth of random geometric graphs in D-dimensional Euclidean space:
- Diversity of sublinear growth — area size $V(t) \sim N(t)^{d/d+1}$
- Superlinear growth of interactions (GDP) — number of edges $E(t) \sim N(t)^{d+2/d+1}$

a b





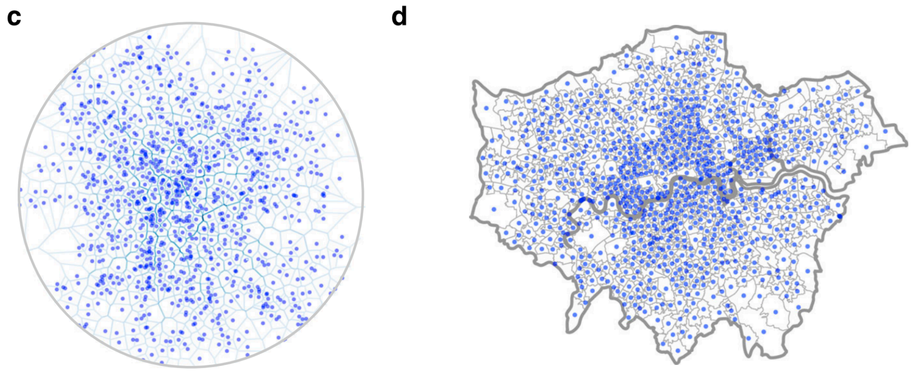


Table 1 Theoretical and empirical results for scaling laws

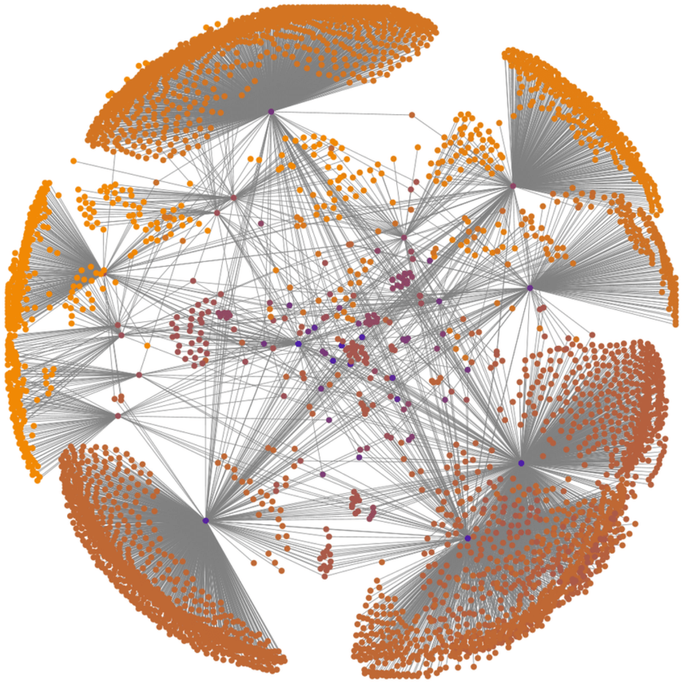
Variables	Theoretical	Empirical[a]
Road length	5/6	(0.74, 0.92)
GDP	7/6	(1.01, 1.33)
Area	2/3	(0.56, 1.04)[b]

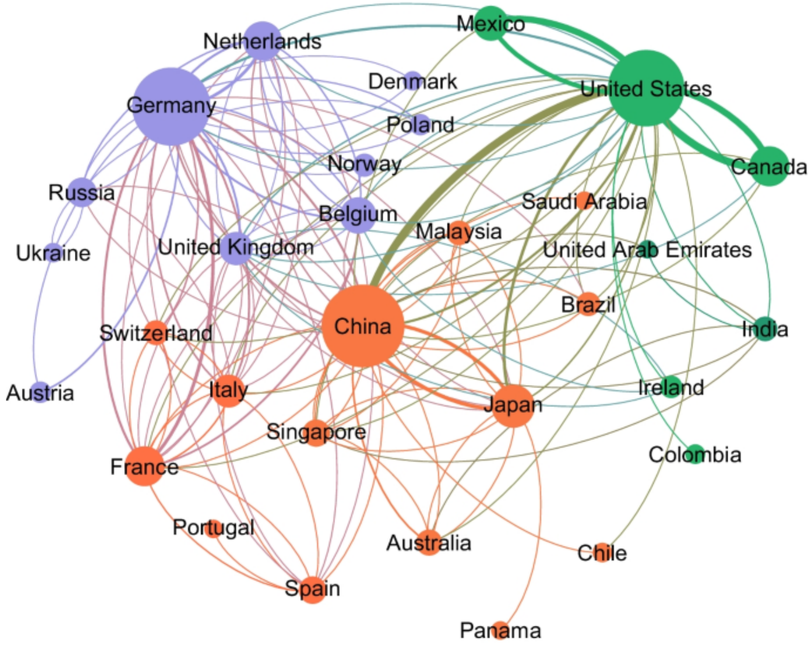
[a]Data from ref.[33]
[b]Based on different definitions of city area, the exponents are quite controversial[25]

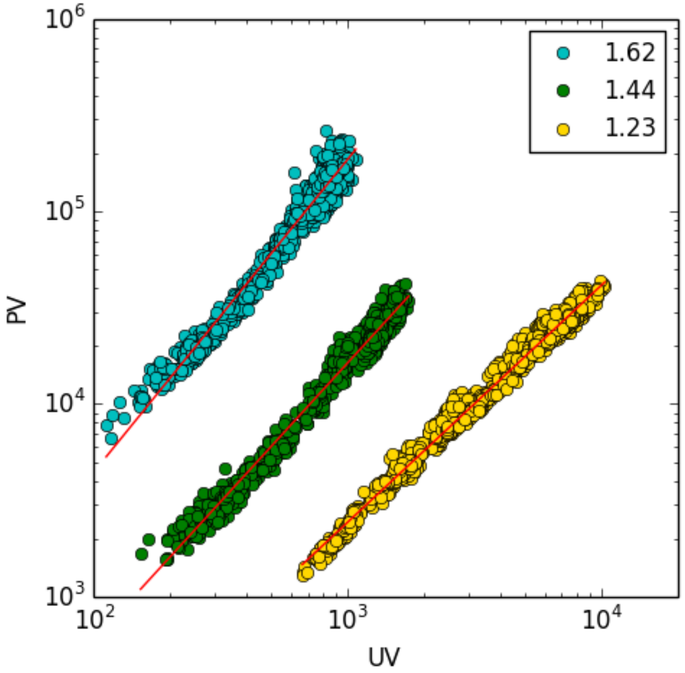
Zhang, J., et al. (2015). Scaling behaviours in the growth of networked systems and their geometric origins. *Scientific Reports*, 5, 1–5.

Li, R., et al. (2017). Simple spatial scaling rules behind complex cities. Nature Communications, 8(1), 1–7.

Potential Universal Mechanism — Extension to Non-Euclidean Spaces

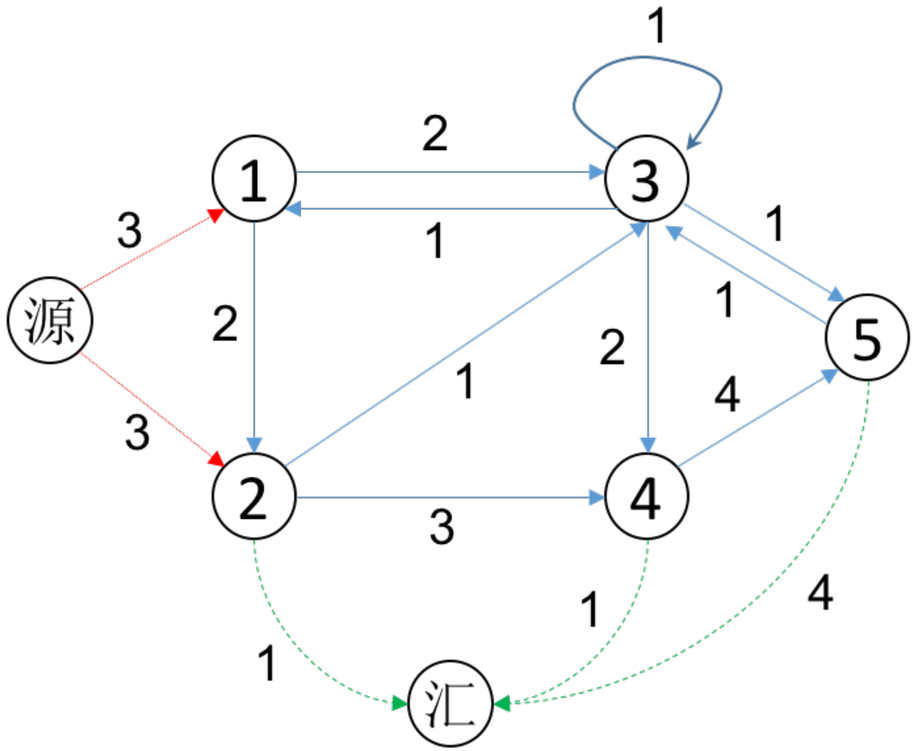






Wu, L., Zhang, J., & Zhao, M. (2014). The Metabolism and Growth of Web Forums. 9(8),

Potential Universal Mechanism — Flow Models



开放流网络

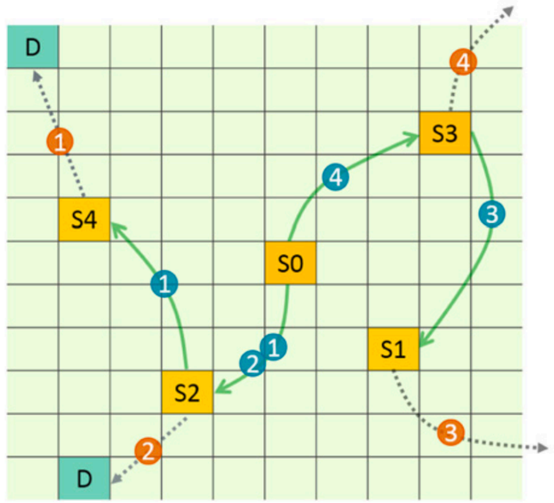
Allometric Scaling Laws in Open Flow Networks

- Node capacity and node flow: $C_i \sim T_i^{\eta}$, where C is node capacity and T is node flow.
- Total system flow and total system inflow: $TST(t) \sim IS(t)^{\theta}$,
  - where TST is total system throughput $TST = \sum T_i = \sum f_{ij'}$
  - and IS is total inflow from the source to node i, $IS = \sum f_{0i}$

Zhang, J., Lou, X., & Guo, L. (2016). Universal patterns and constructal law in open flow networks. International Journal of Heat and Technology, 34(1), S75–S82.

Potential Universal Mechanism — Flow Models

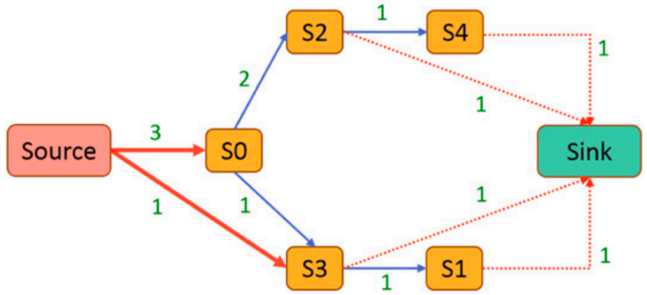
a.

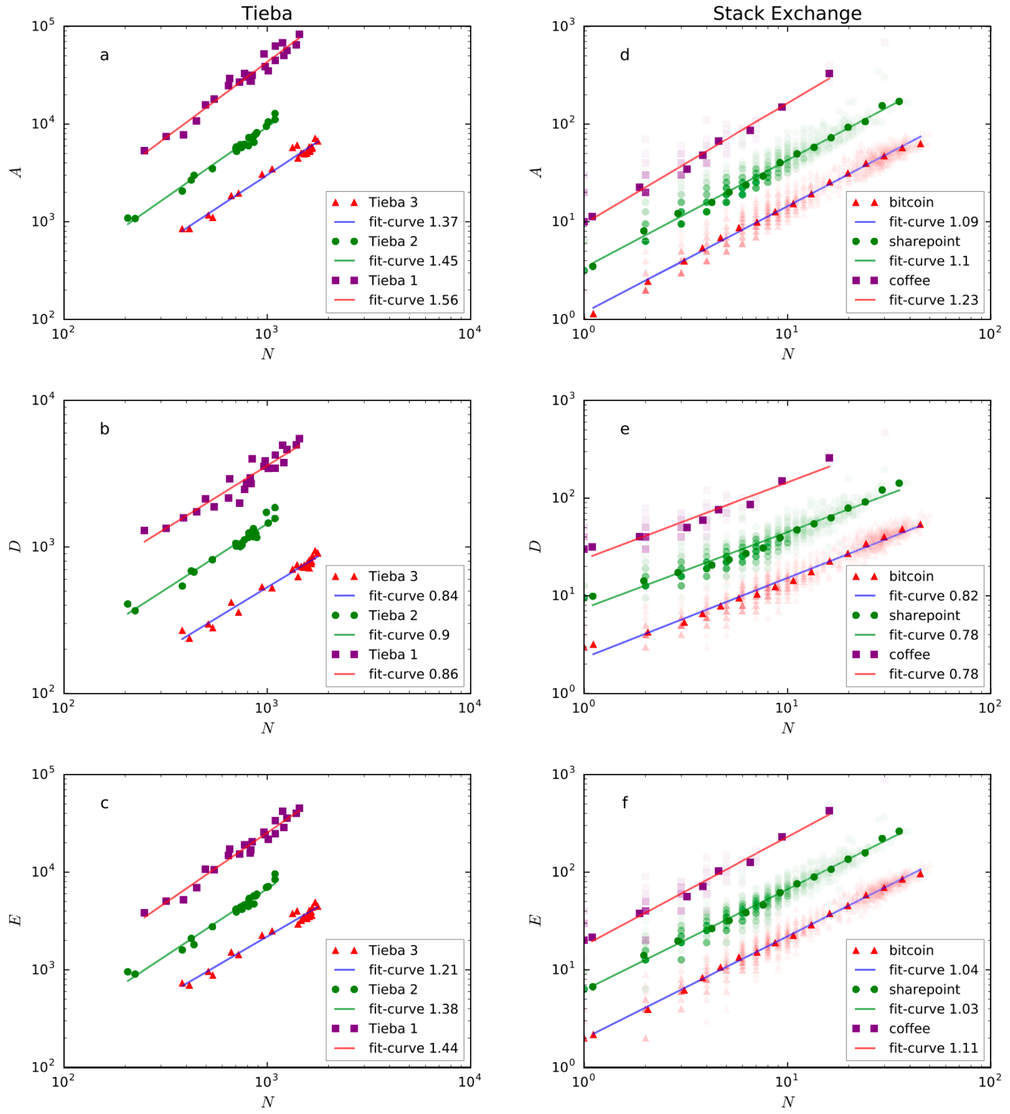


b.

Active Site:	User:	User's Activities:
S0	#1	S0→S2→S4→D
S1	#2	S0→S2→D
S2	#3	S3→S1→D
S3	#4	S0→S3→D
S4		

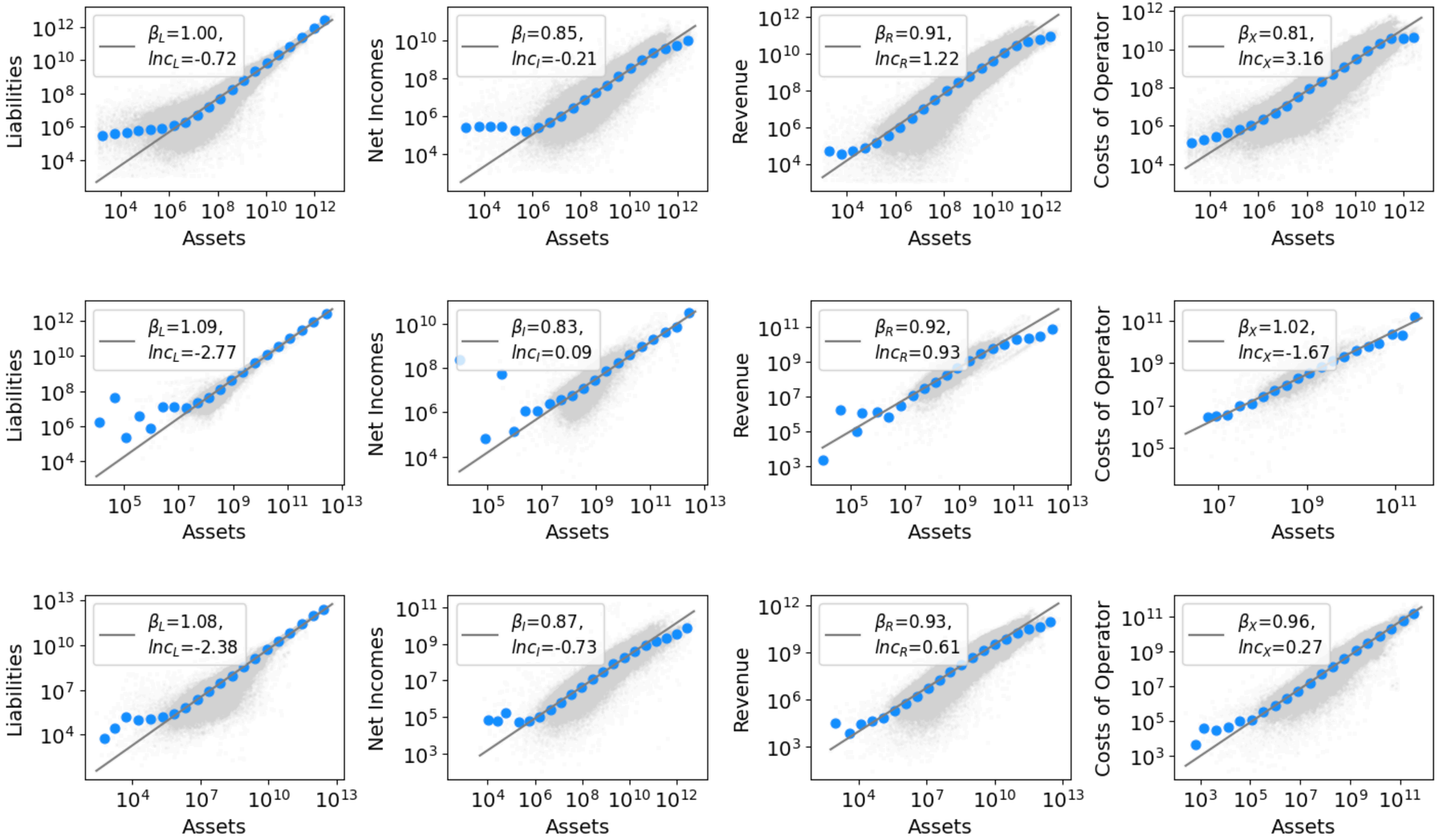
c.



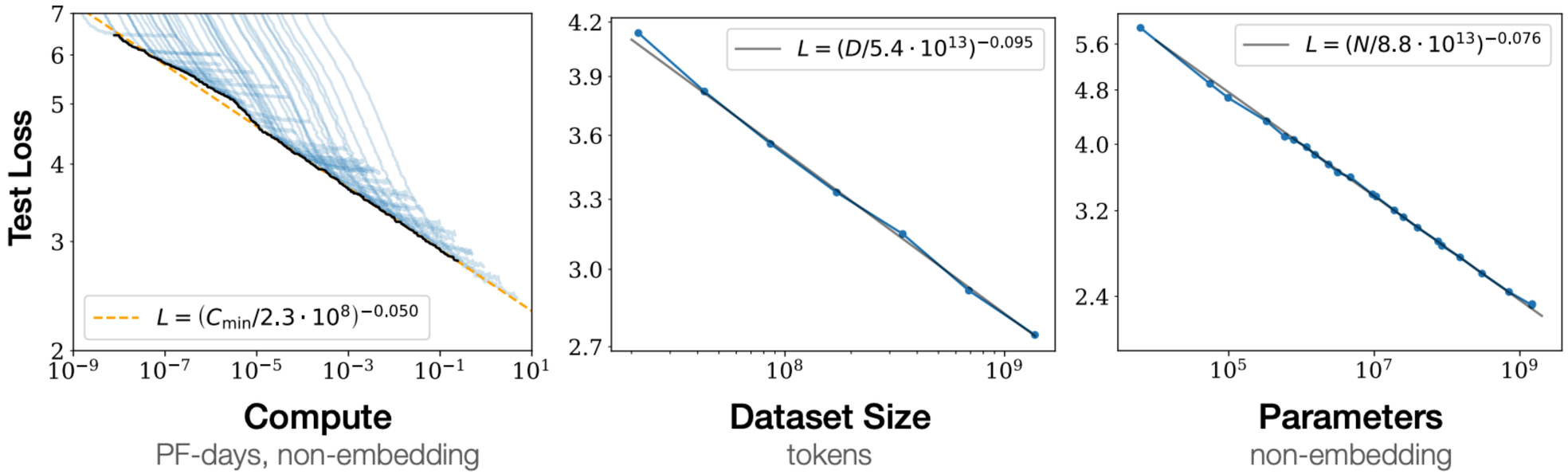


Zeng, F., Gong, L., Liu, J., Zhang, J., Chen, Q., & Xin, R. (2020). Human mobility in interest space and interactive random walk. Journal of Physics: Complexity, 1(2).

Inspirations for More Complex Systems



Inspirations for More Complex Systems



The Quantization Model of Neural Scaling, 2024

Summary

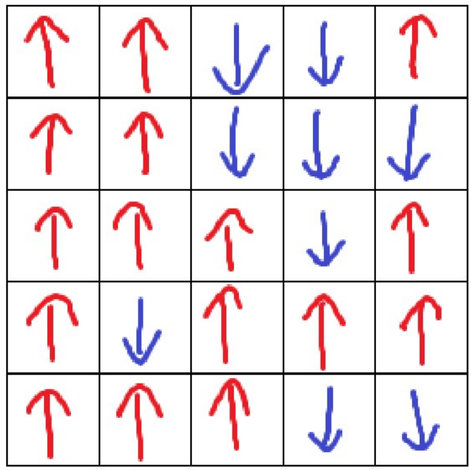
- For complex systems, scaling laws imply self-similarity in system structure.
- Santa Fe researchers represented by West and Bettencourt pioneered complex-system modeling theories based on fractal structures, with influential applications in biology and urban systems.
- Flow models exhibit broader modeling power due to their simplicity.
- Open flow network theory, extended from flow models, is a powerful tool for modeling scaling laws in non-Euclidean systems, though more general growth mechanisms still require further study.

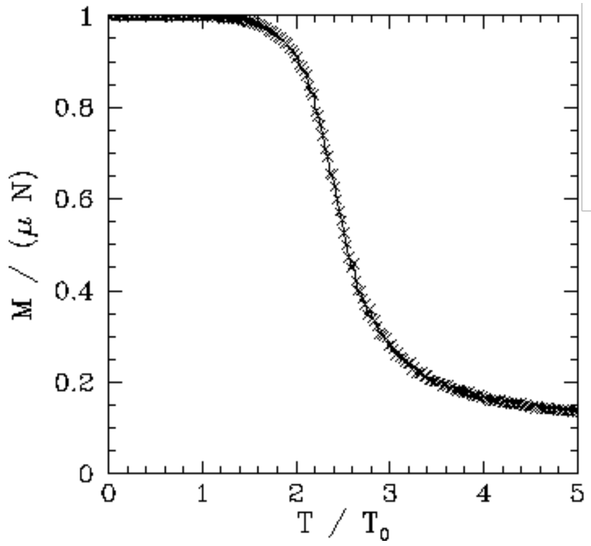
- What is Fractal — Self-Similarity
  - Origin of Fractals: Mathematics Monsters
  - Definition and Quantification of Fractals
    - Definition Proposed by Mandelbrot
    - Proposition and Calculation of Fractal Dimensions

- "Fractals" in Complex Systems — Scaling Theory
  - Kleiber's Law and Its Origin
  - Urban Scaling Laws and Their Origin
  - Underlying Universal Mechanisms

- Broader Discussion of Self-Similarity in Complex Systems
  - Critical Phase Transition and Self-Similarity
  - Self-Similarity in Dynamics
  - Self-Similarity in Complex Networks

Critical Phase Transitions and Self-Similarity — Criticality

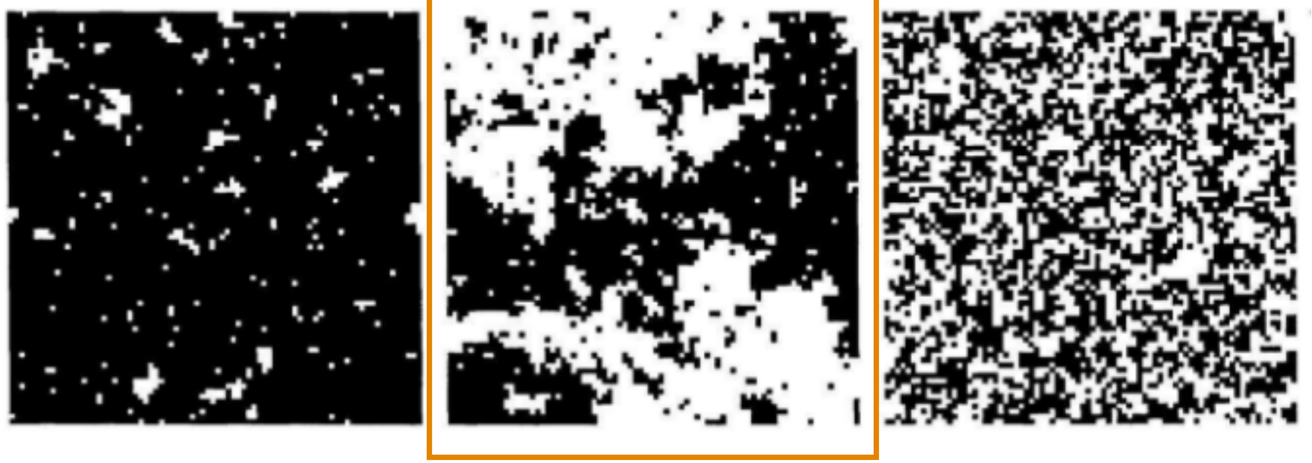




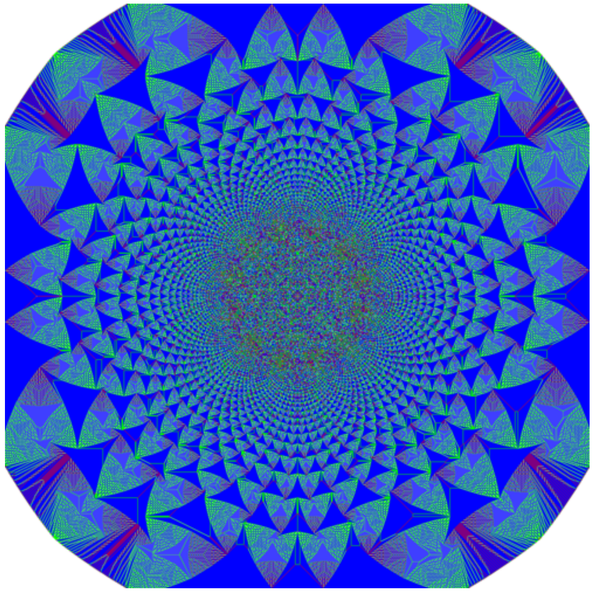
$$H = -\sum_{i,j} J_{ij}\sigma_i\sigma_j + h\sum_i \sigma_i$$

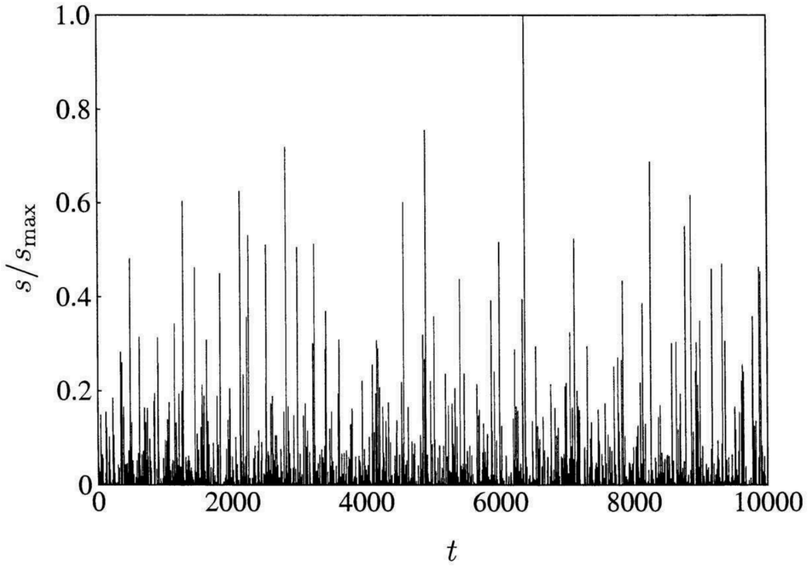
$$M = \sum_i \sigma_i$$

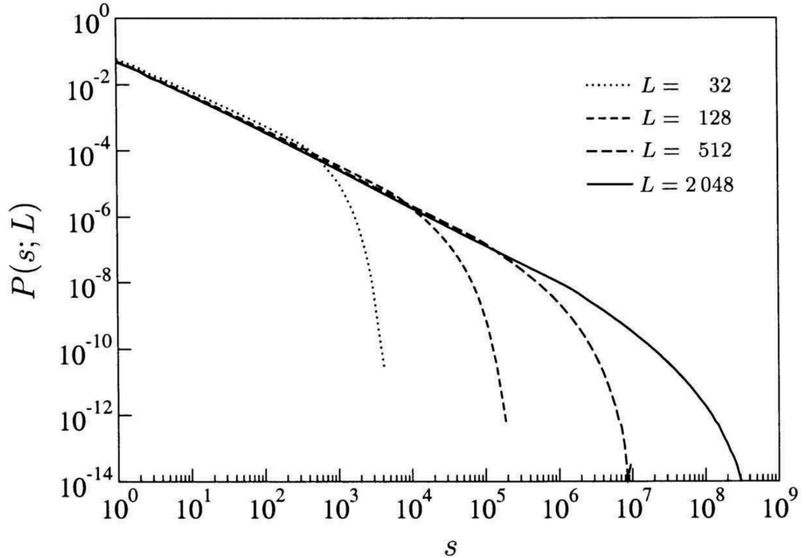
$T < T_c$ $T = T_c$ $T > T_c$

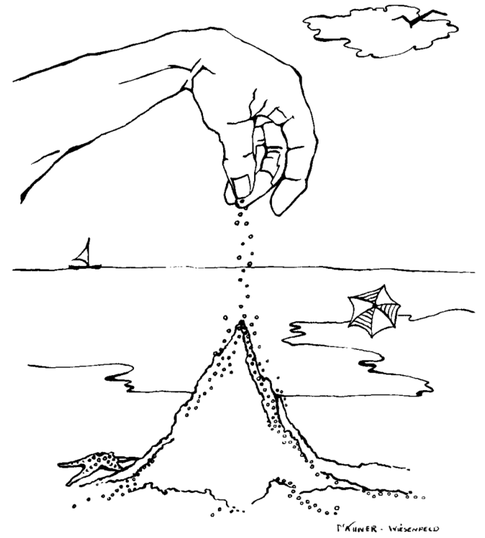


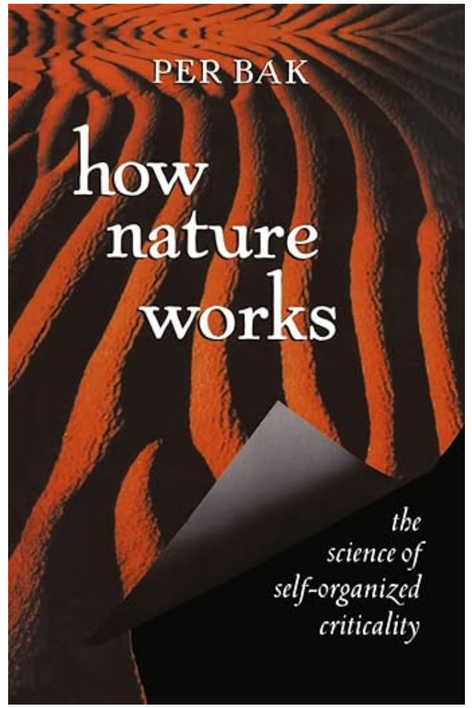
Critical Phase Transitions and Self-Similarity — Self-Organized Criticality



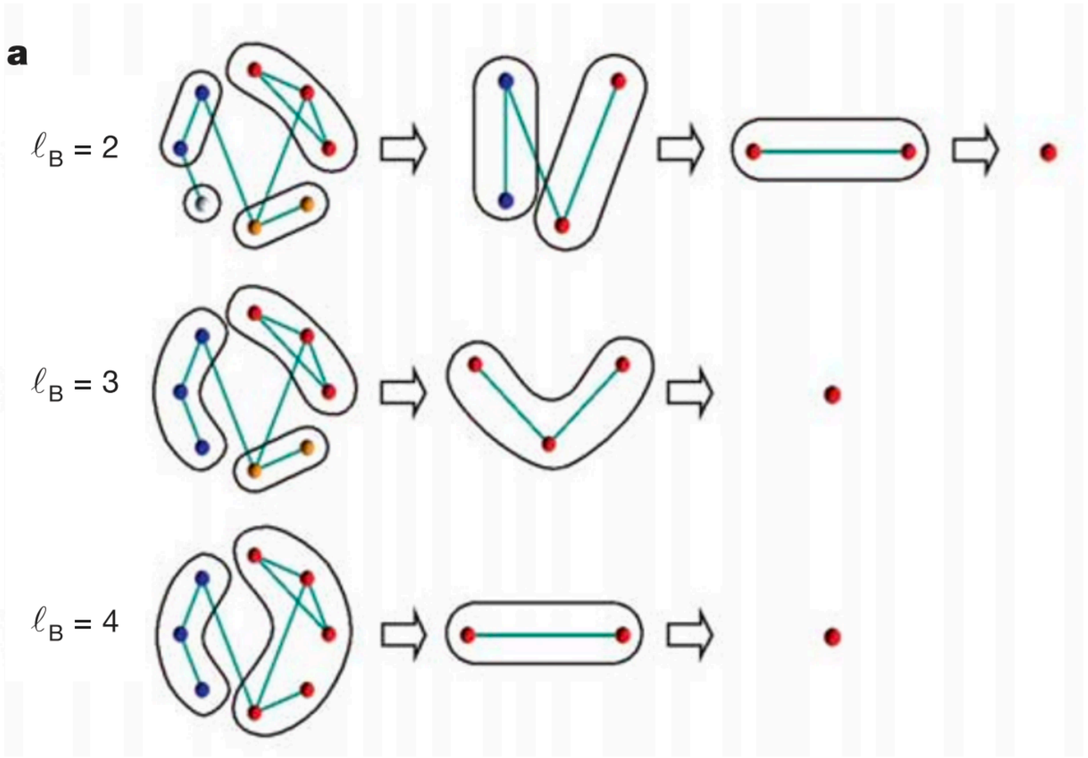


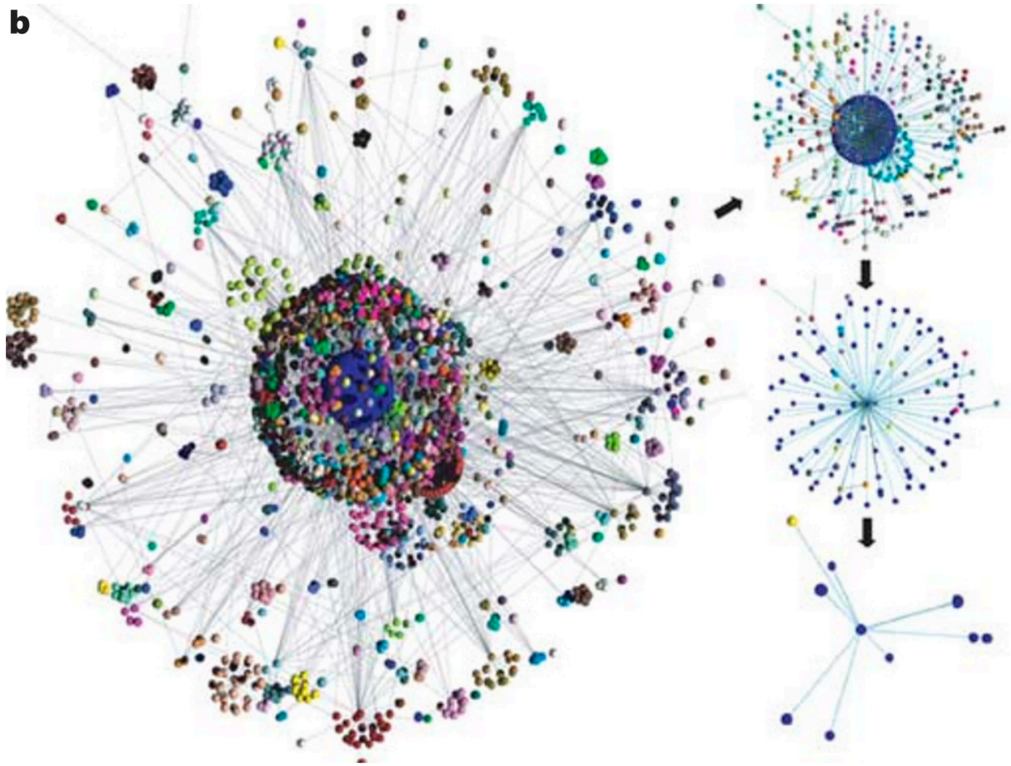






Self-Similarity in Complex Networks





Box-counting on networks: an extension from Euclidean distance to network distance. $N_B \approx \ell_B^{-d_B}$

Song, C., & Havlin, S. (2005). Self-similarity of complex networks. Nature, 433(January), 2–5.

Self-Similarity in Complex Networks

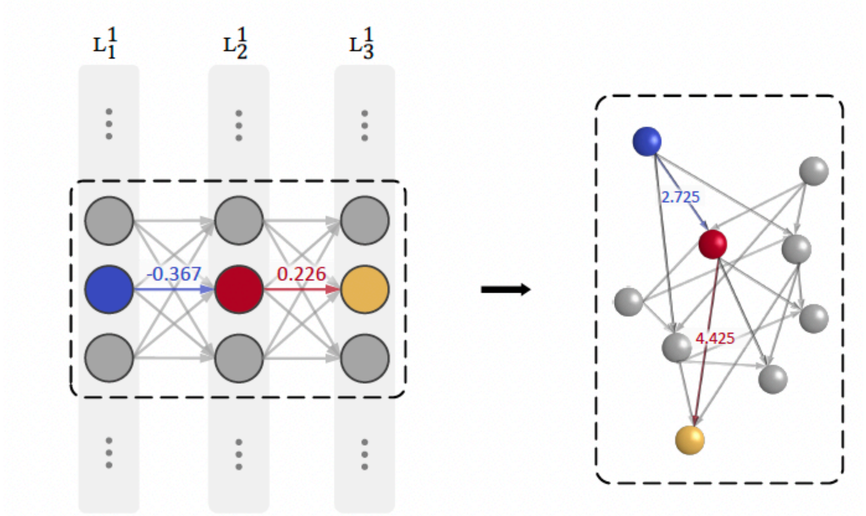
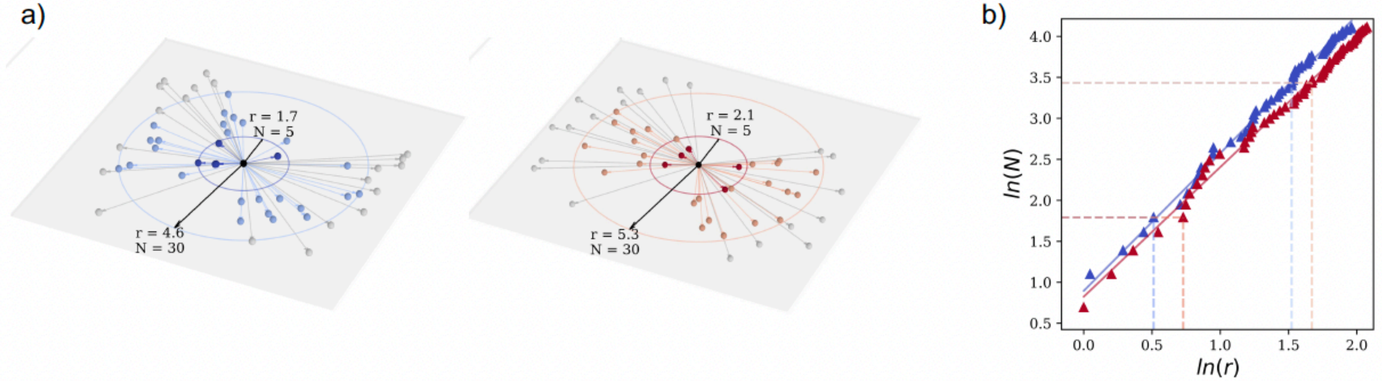


Figure 3: The transformation from a partial neural network (NN) in large models into a neuron interaction network (NIN).



Neuron-based Multifractal Analysis of Neuron Interaction Dynamics in Large Models,2025,ICLR

Renormalization of Complex Networks

nature reviews physics

https://doi.org/10.1038/s42254-025-00817-5

Technical review

Check for updates

Network renormalization

Andrea Gabrielli[1,2,3], Diego Garlaschelli[4,5] ✉, Subodh P. Patil[5] & M. Ángeles Serrano[6,7,8]

Abstract

The renormalization group (RG) is a powerful theoretical framework. It is used on systems with many degrees of freedom to transform the description of their configurations, along with the associated model parameters and coupling constants, across different levels of resolution. The RG also provides a way to identify critical points of phase transitions and study the system's behaviour around them. In traditional physical applications, the RG largely builds on the notions of homogeneity, symmetry, geometry and locality to define metric distances, scale transformations and self-similar coarse-graining schemes. More recently, efforts have been made to extend RG concepts to complex networks. However, in such systems, explicit geometric coordinates do not necessarily exist, different nodes and subgraphs can have different statistical properties, and homogeneous lattice-like symmetries are absent — all features that make it complicated to define consistent renormalization procedures. In this Technical Review, we discuss the main approaches, important advances, and the remaining open challenges for network renormalization.

Sections

Introduction

Challenges for renormalization of complex networks

Network coarse-graining approaches

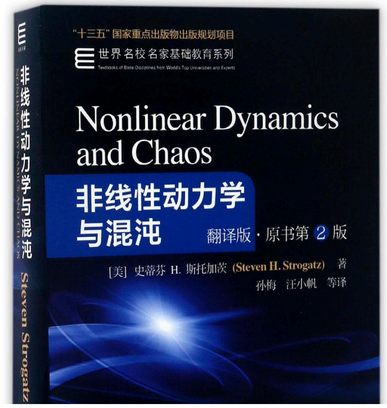
Geometric network renormalization

Laplacian network renormalization

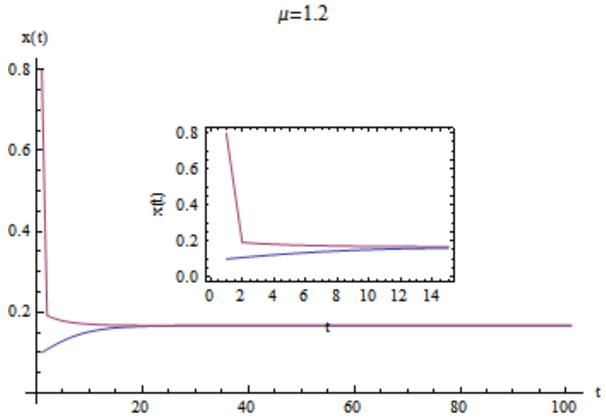
Multiscale network renormalization

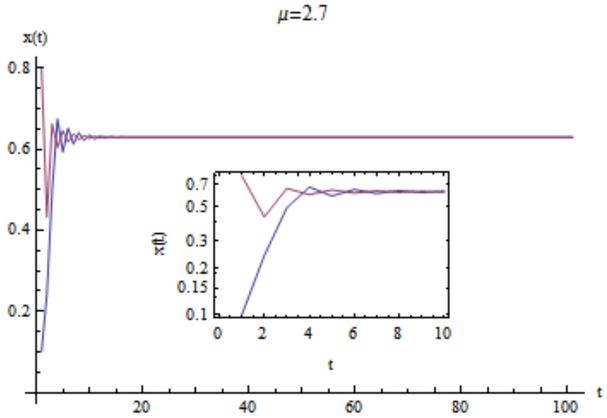
Outlook

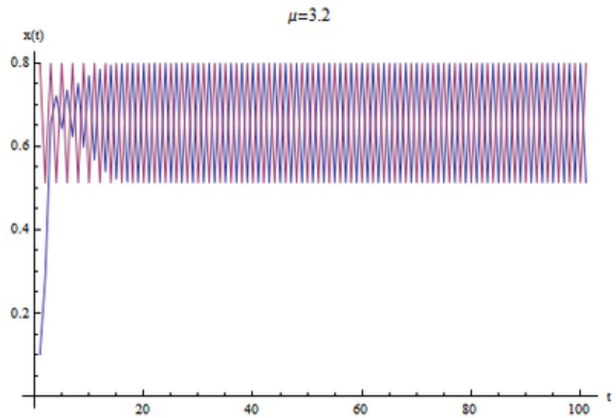
Self-Similarity in Dynamical Systems — Chaotic Dynamics

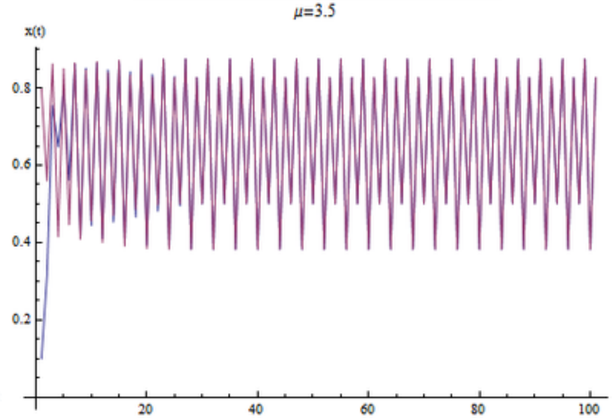


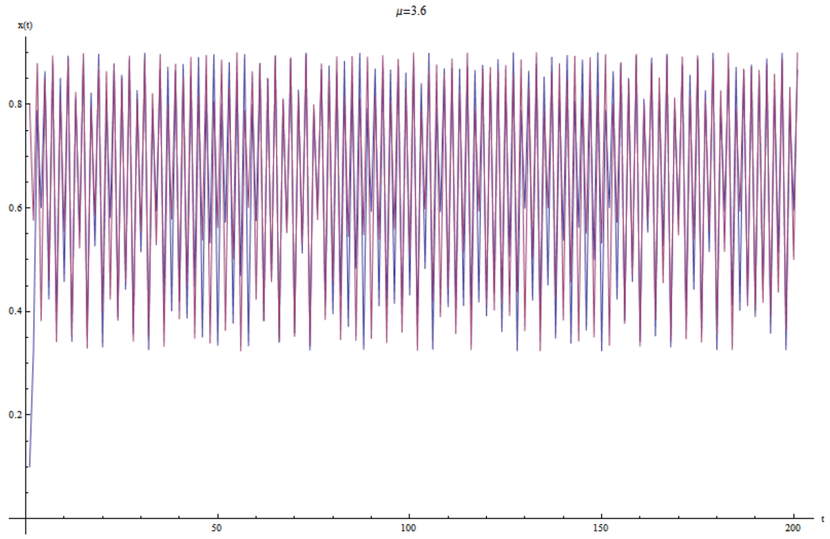
$$x_{t+1} = rx_t(1 - x_t)$$

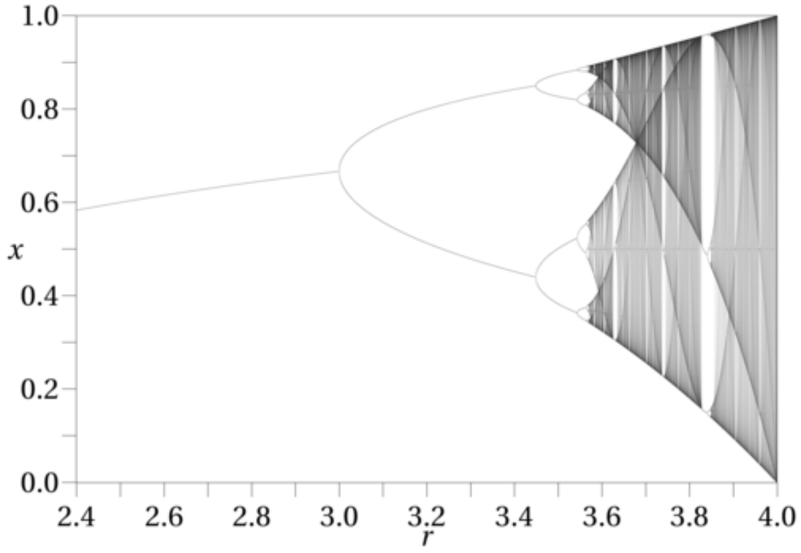






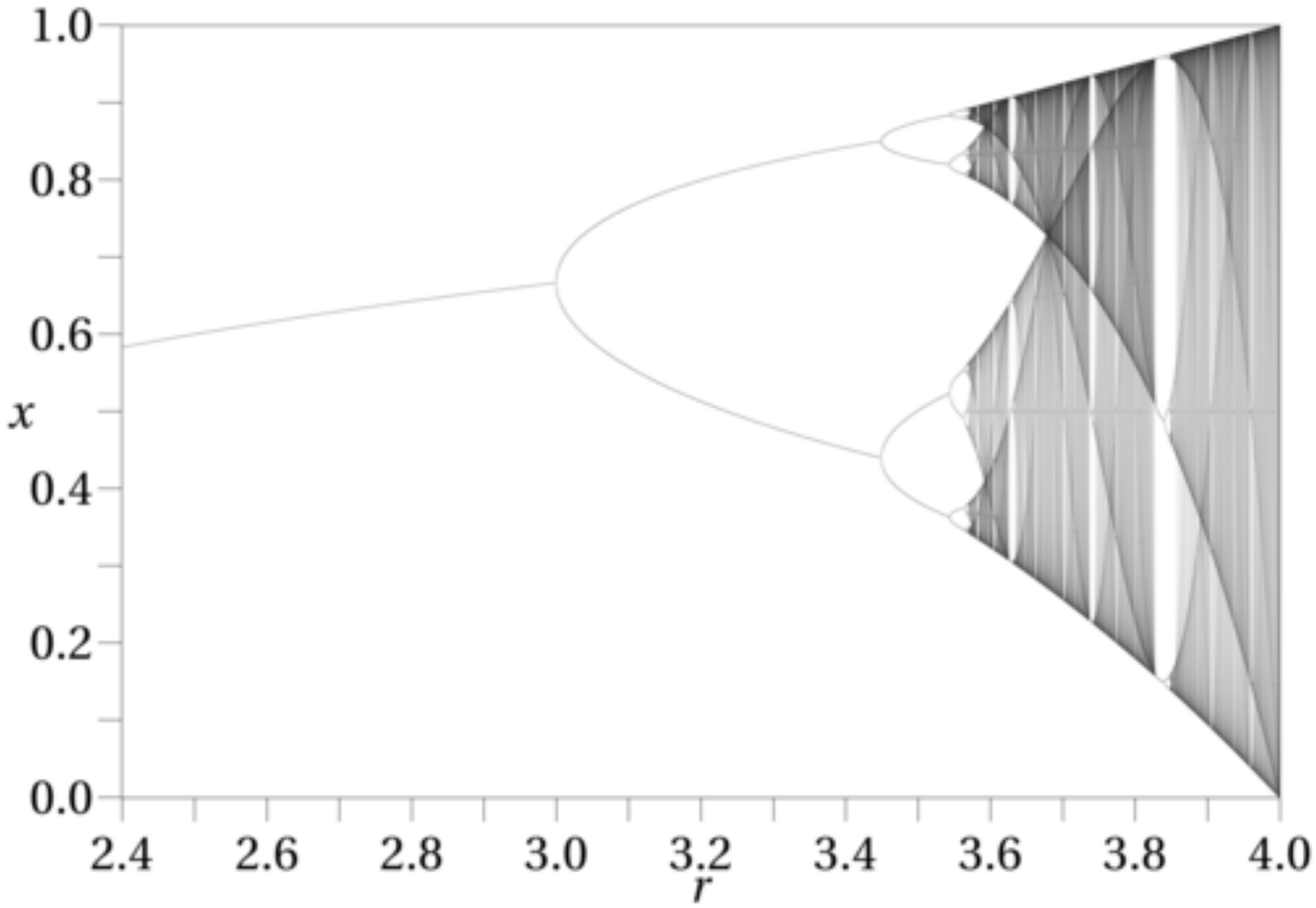






Self-Similarity in Dynamical Systems — Chaotic Dynamics

$$x_{t+1} = r x_t (1 - x_t)$$



- Fractal structure of period-doubling bifurcations (the first Feigenbaum constant)

$$\delta = \lim_{n\to\infty} \frac{a_n - a_{n-1}}{a_{n+1} - a_n} = 4.6692016091029909$$

- Iterative point sets in chaotic states have dimensions between 0 and 1.
- Phase space: fractal dimension of strange attractors.

Summary

- Scale invariance means that every scale is equally important.
- It implies long-range correlations and thus system complexity.
- Universal mechanisms? A long journey still lies ahead.